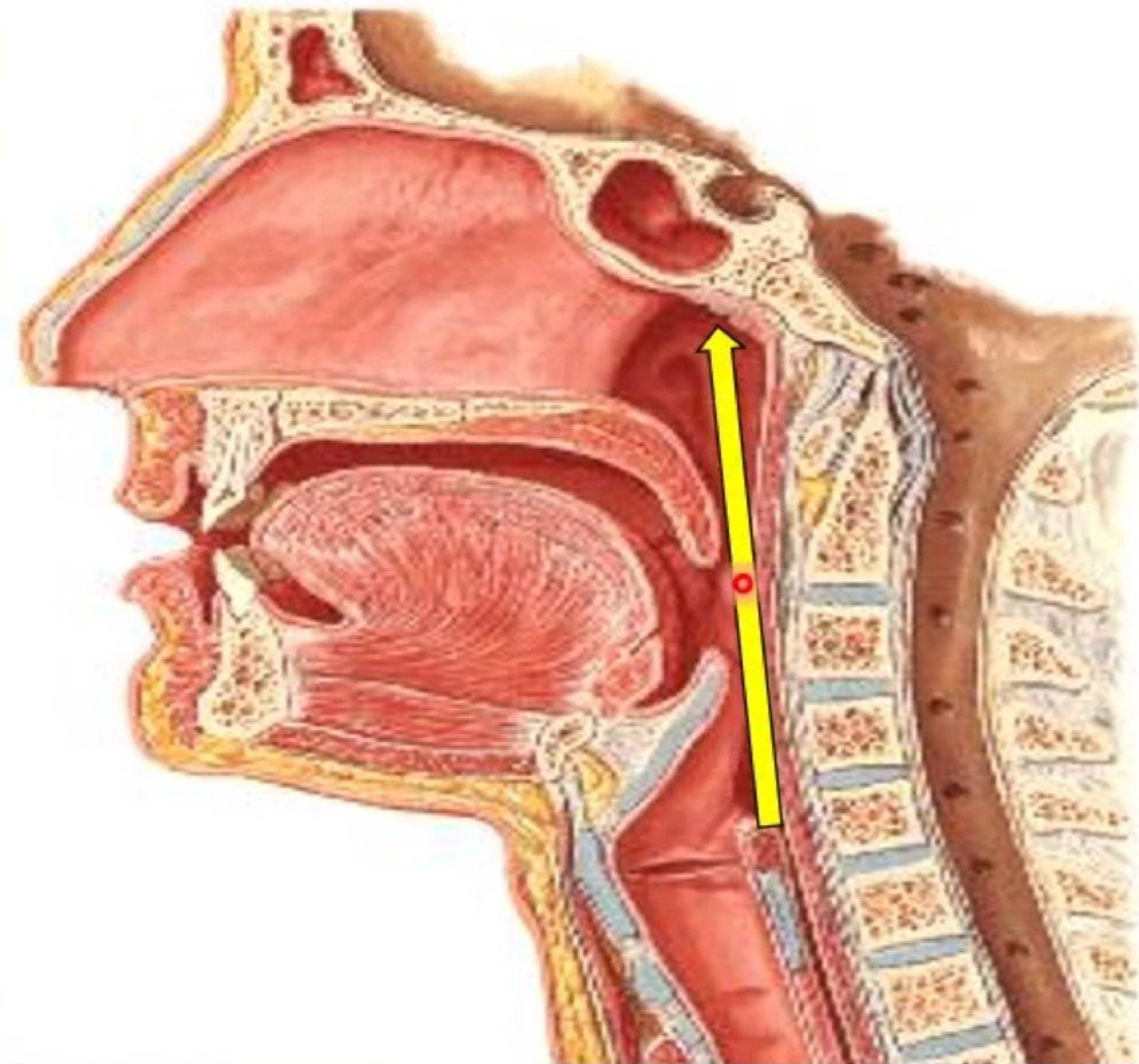


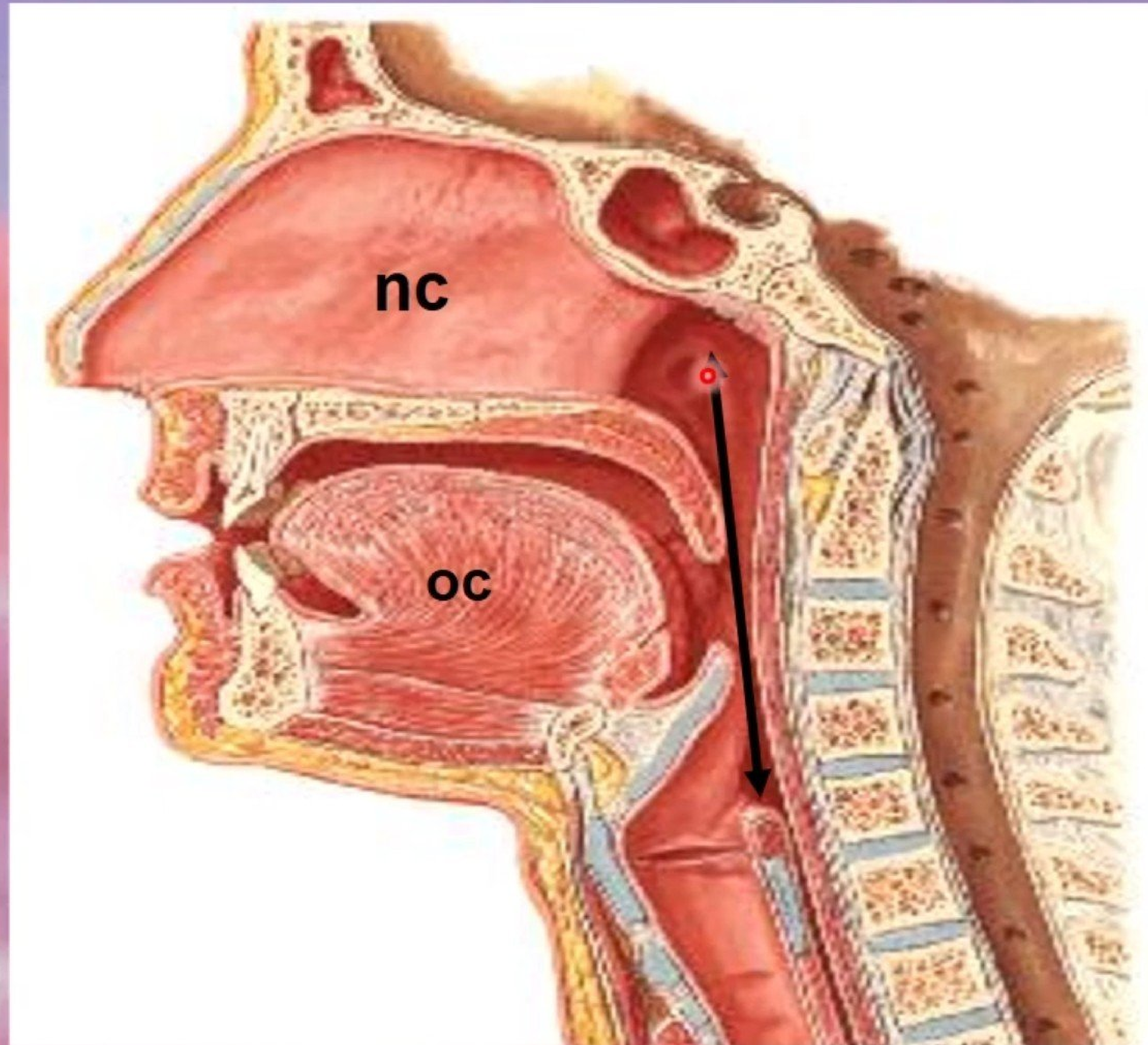
Pharynx

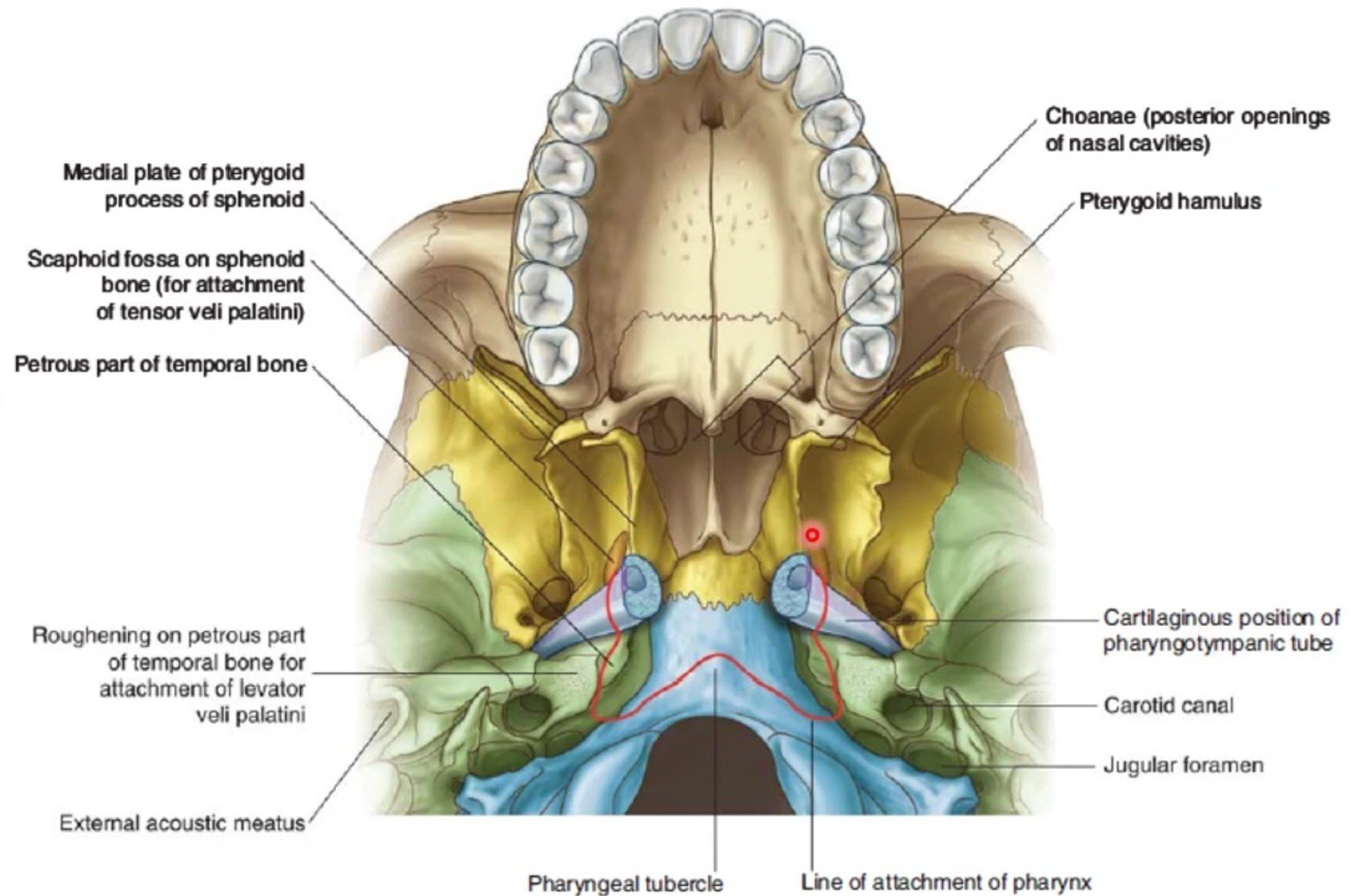


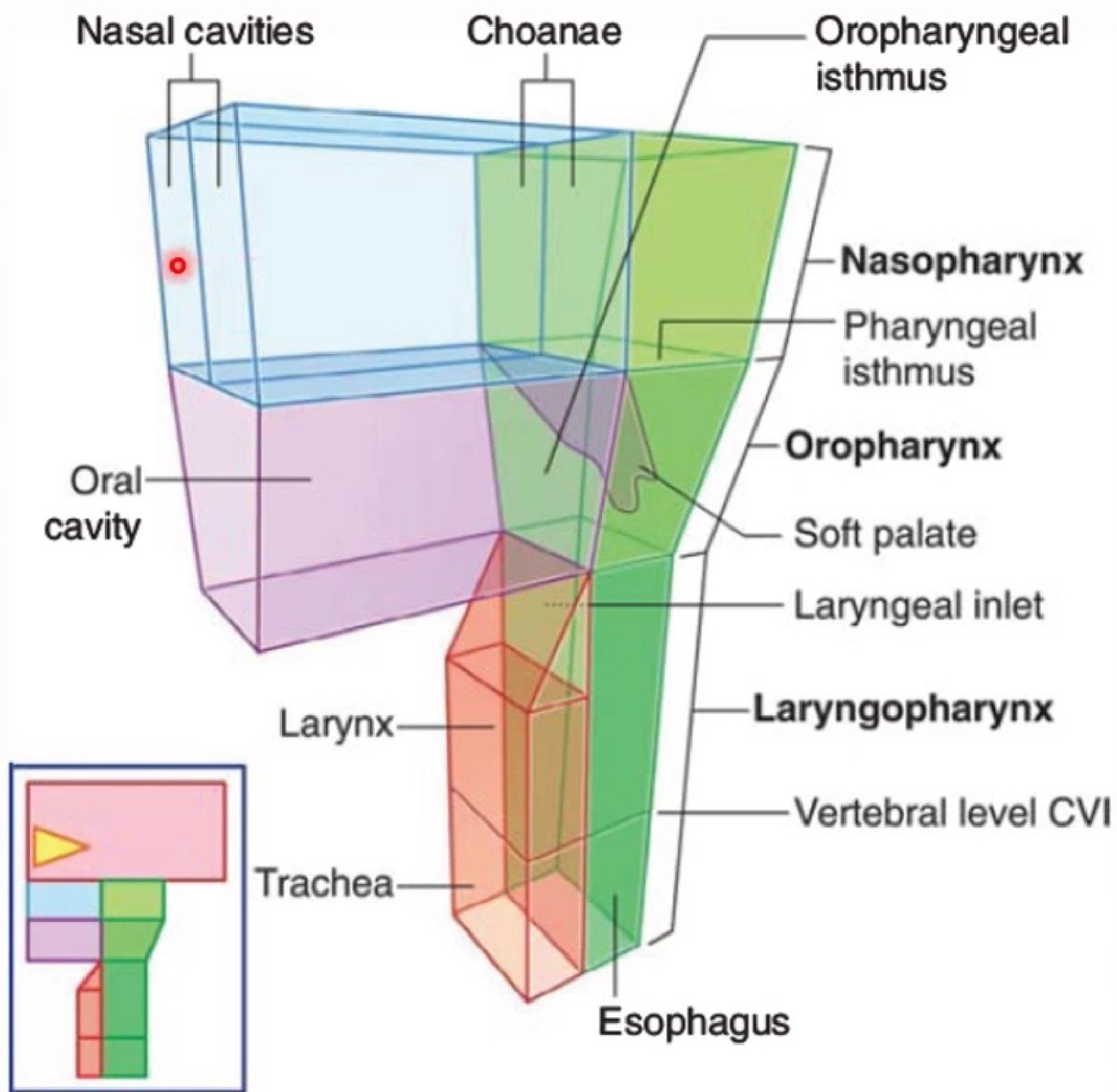
Dr. Osman Ahmed Osman
Department of Anatomy

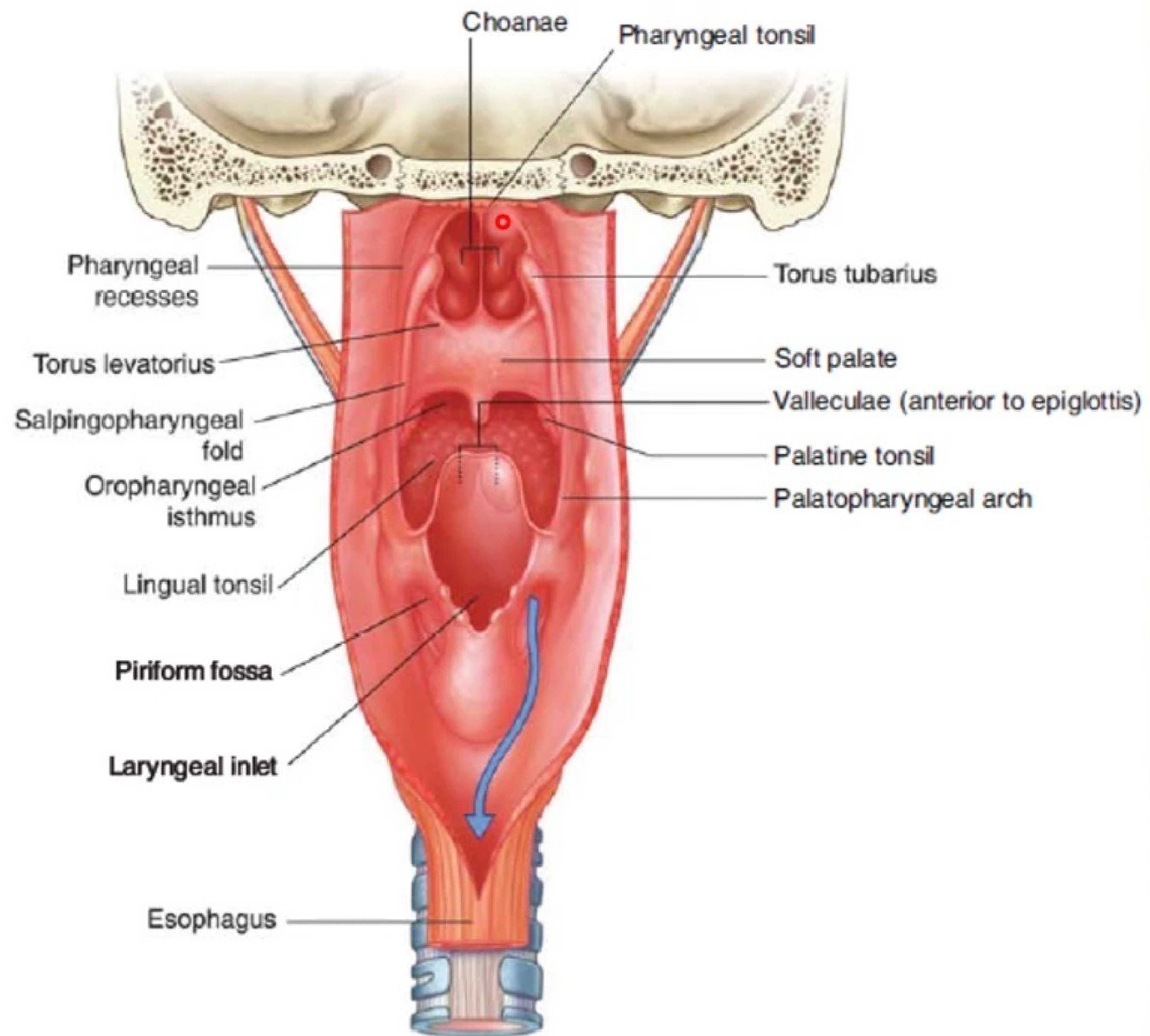
Pharynx

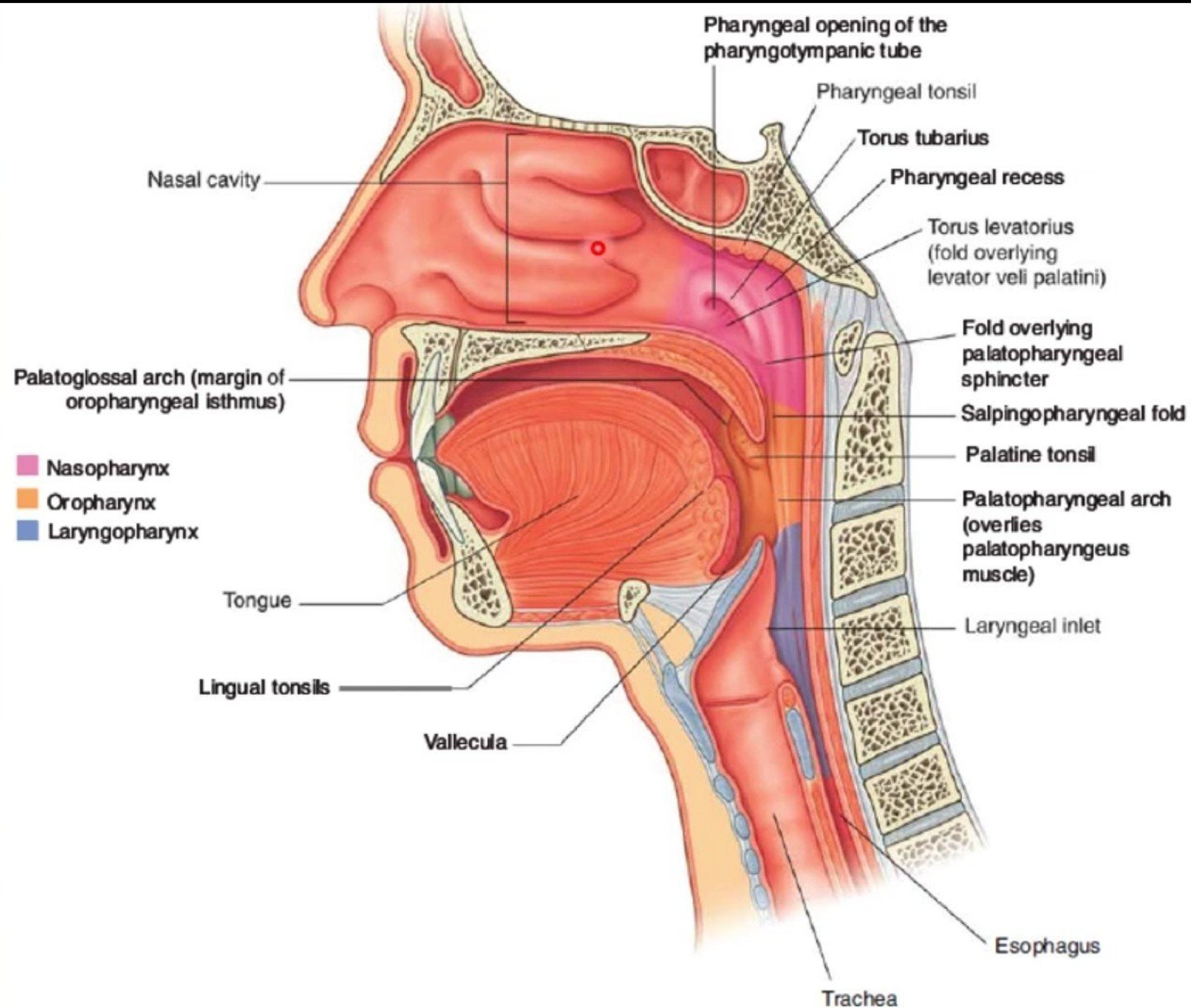
- Muscular tube lying behind the **nose, oral cavity & larynx**.
- Extends **from the base of the skull to level of the 6th cervical vertebra**, where it is continuous with the esophagus
- The anterior wall is deficient and shows (from above downward):
 - **Posterior nasal apertures.**
 - **Opening of the oral cavity.**
 - **Laryngeal inlet.**
- The muscles arranged in **circular and longitudinal layers**

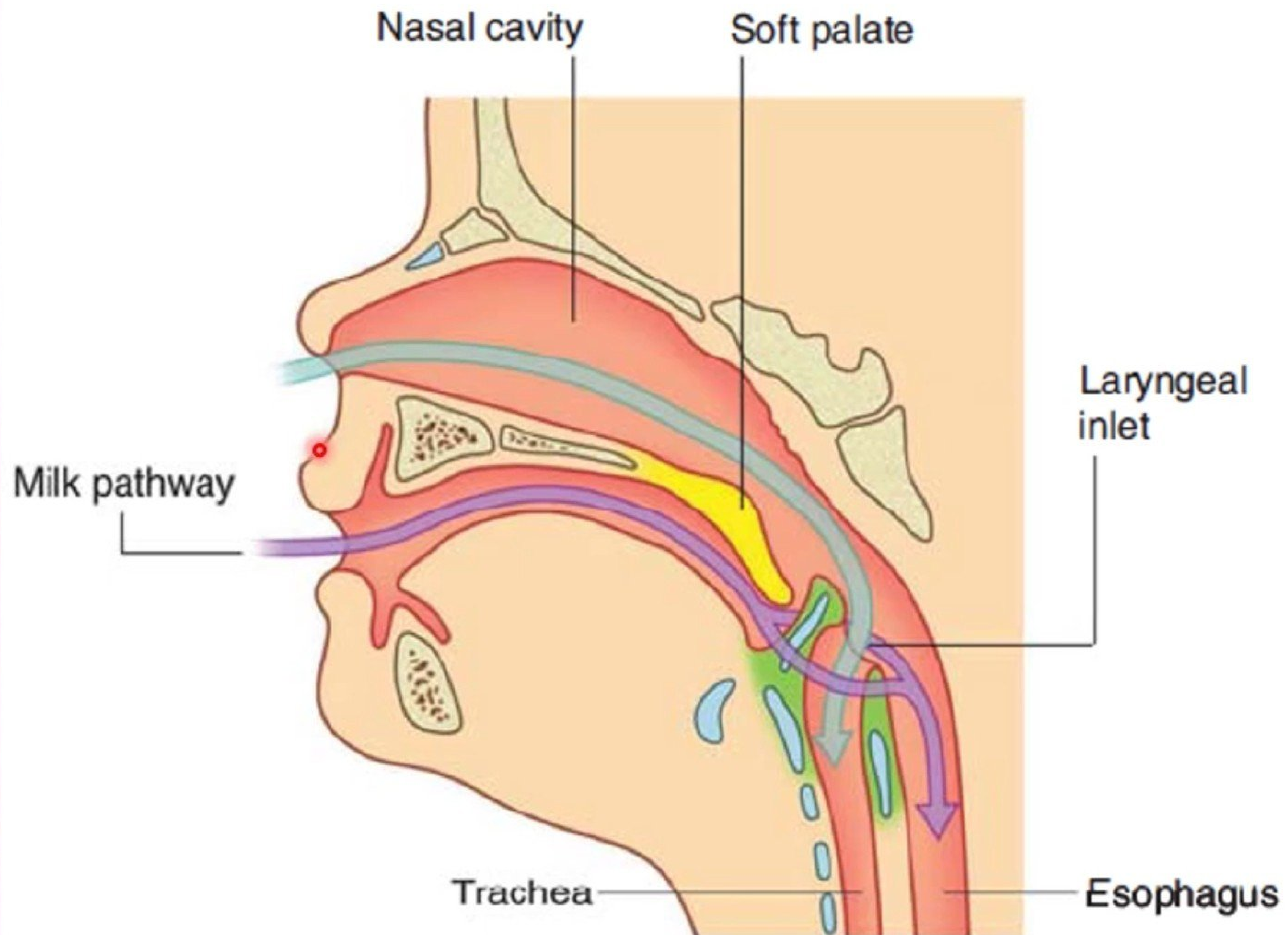










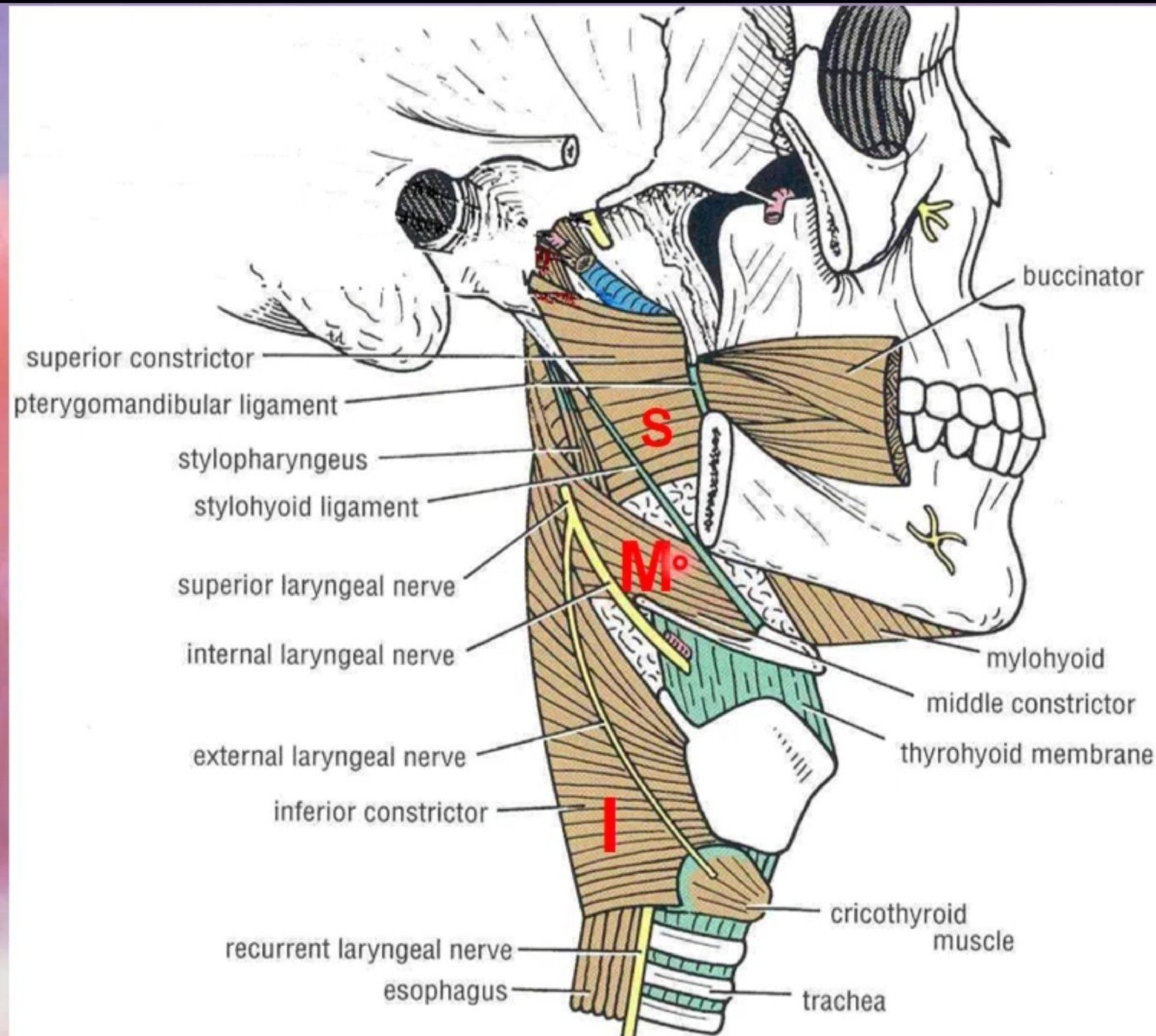


Circular (Constrictor) Muscles

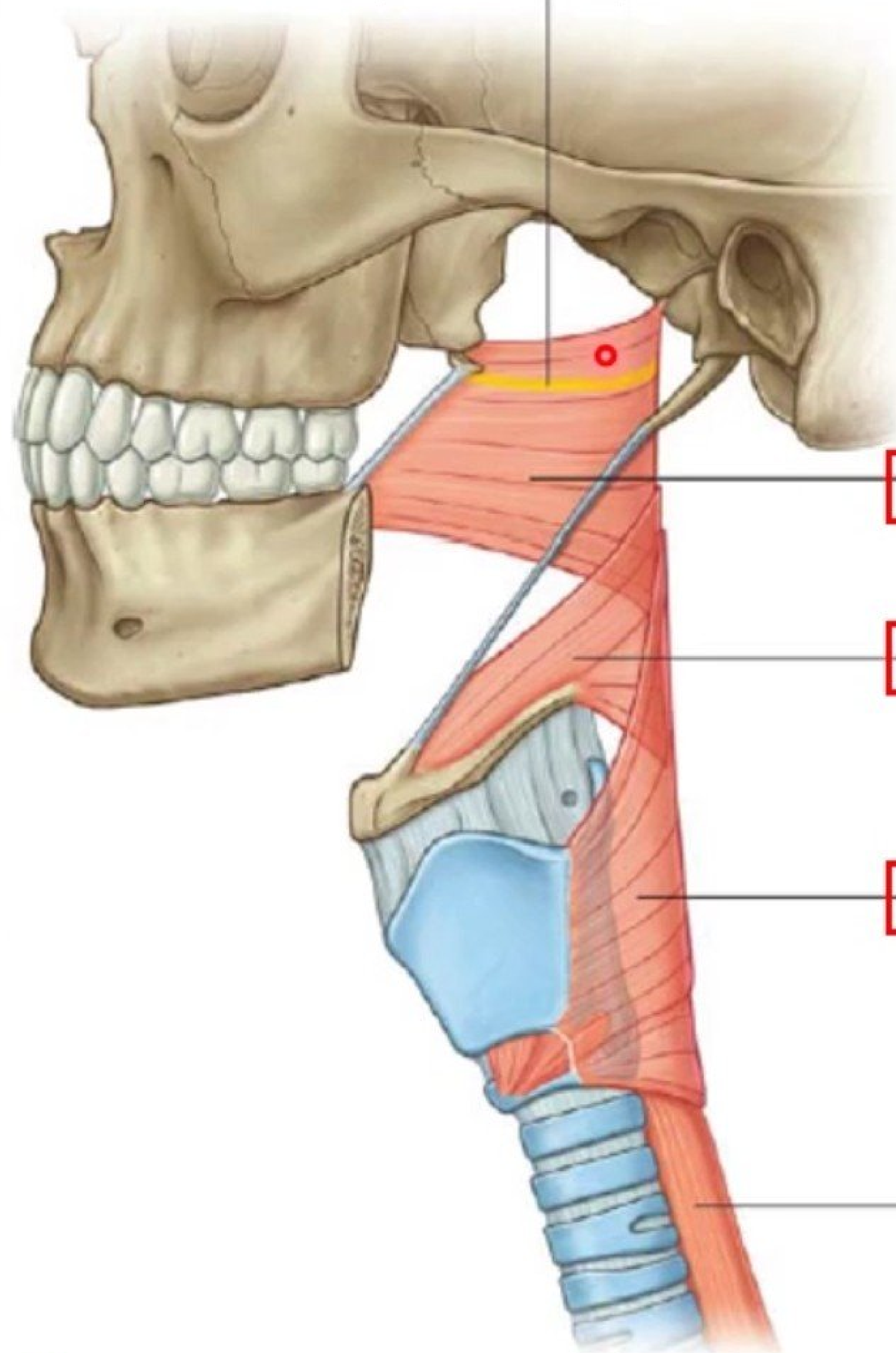
- Three in number:
- **Superior constrictor,**
- **Middle constrictor &**
- **Inferior constrictor**
- The three muscles overlap each other.

Functions:

- Propel the bolus of food down into the esophagus.
- lower fibers of the inferior constrictor (**Cricopharygeus**) act as a sphincter, preventing the entry of air into the esophagus between the acts of swallowing.



Position of palatopharyngeal sphincter
on deep surface of superior constrictor



A

Pharyngeal fascia

Pharyngeal
tubercle

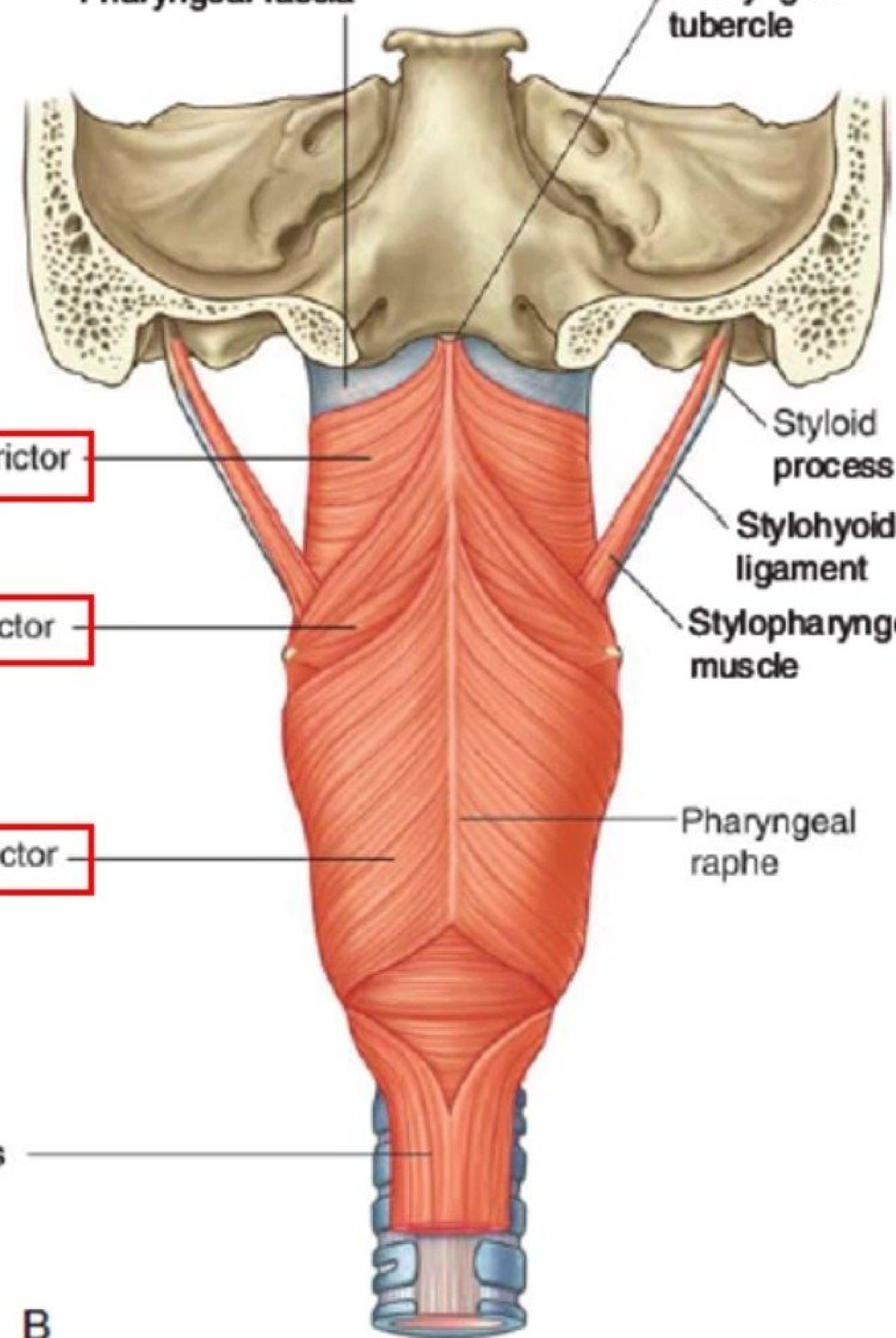
Superior constrictor

Middle constrictor

Inferior constrictor

Esophagus

B



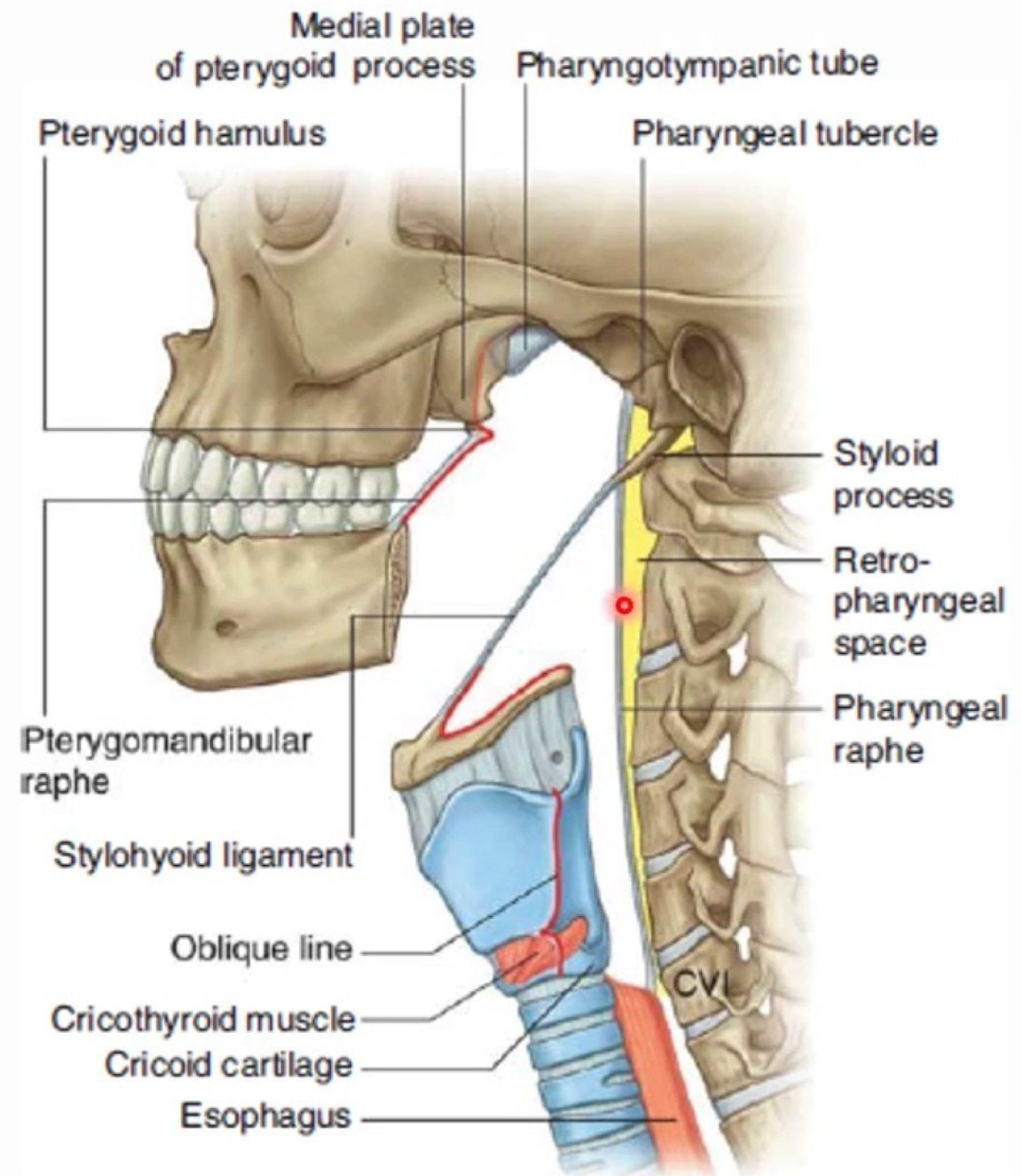
Styloid
process

Stylohyoid
ligament

Stylopharyngeus
muscle

Pharyngeal
raphe

Attachment of lateral pharyngeal wall



The gap between the superior and middle constrictors:

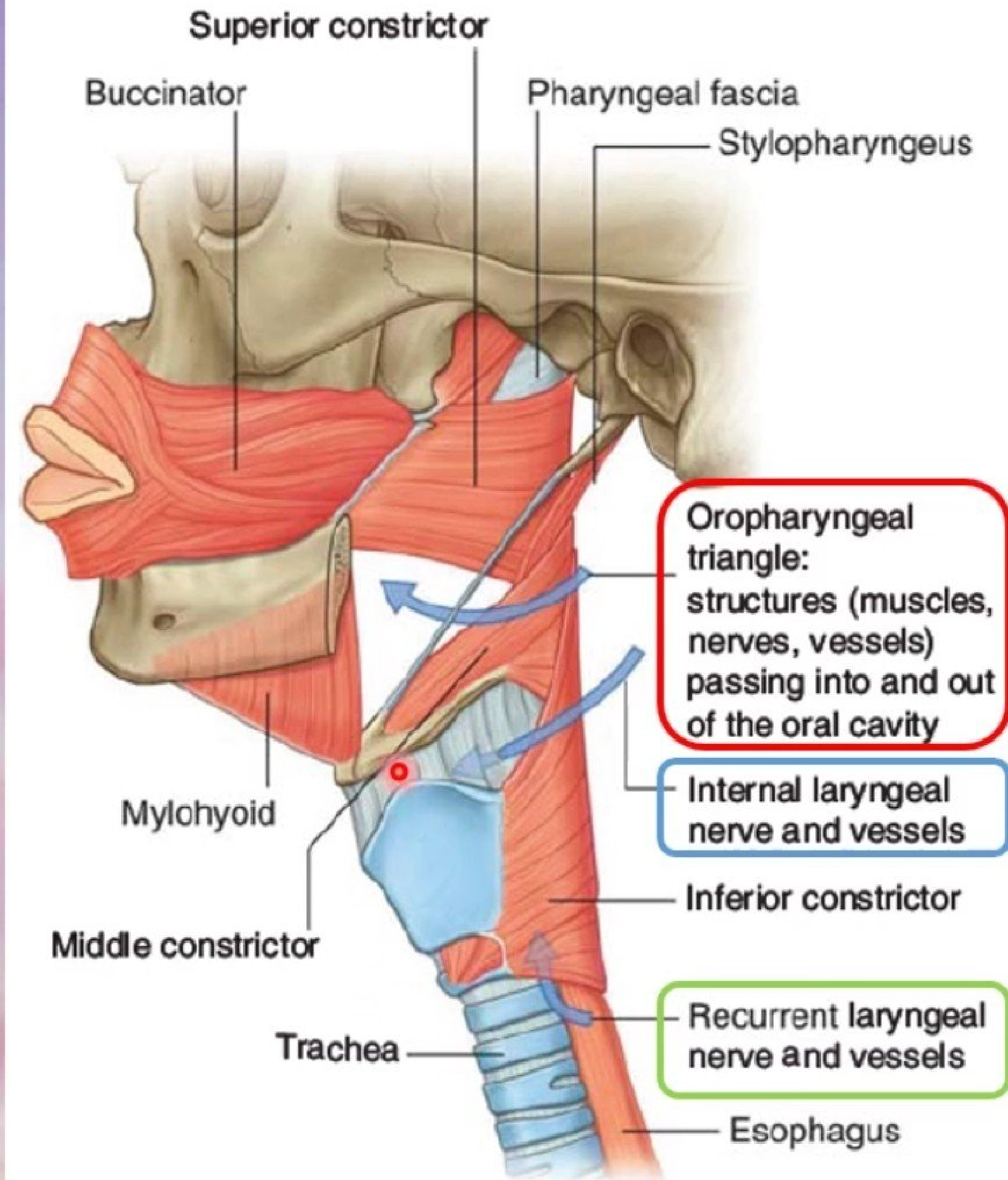
- Stylopharyngeus passes down into the pharynx.
- Styloglossus.
- Glossopharyngeal to the tongue.
- lingual nerves pass to the tongue.

The anterior gap between the middle and inferior constrictors is closed by the thyrohyoid membrane in this gap by piercing the membrane are:

- The internal laryngeal nerve.
- The superior laryngeal vessels.

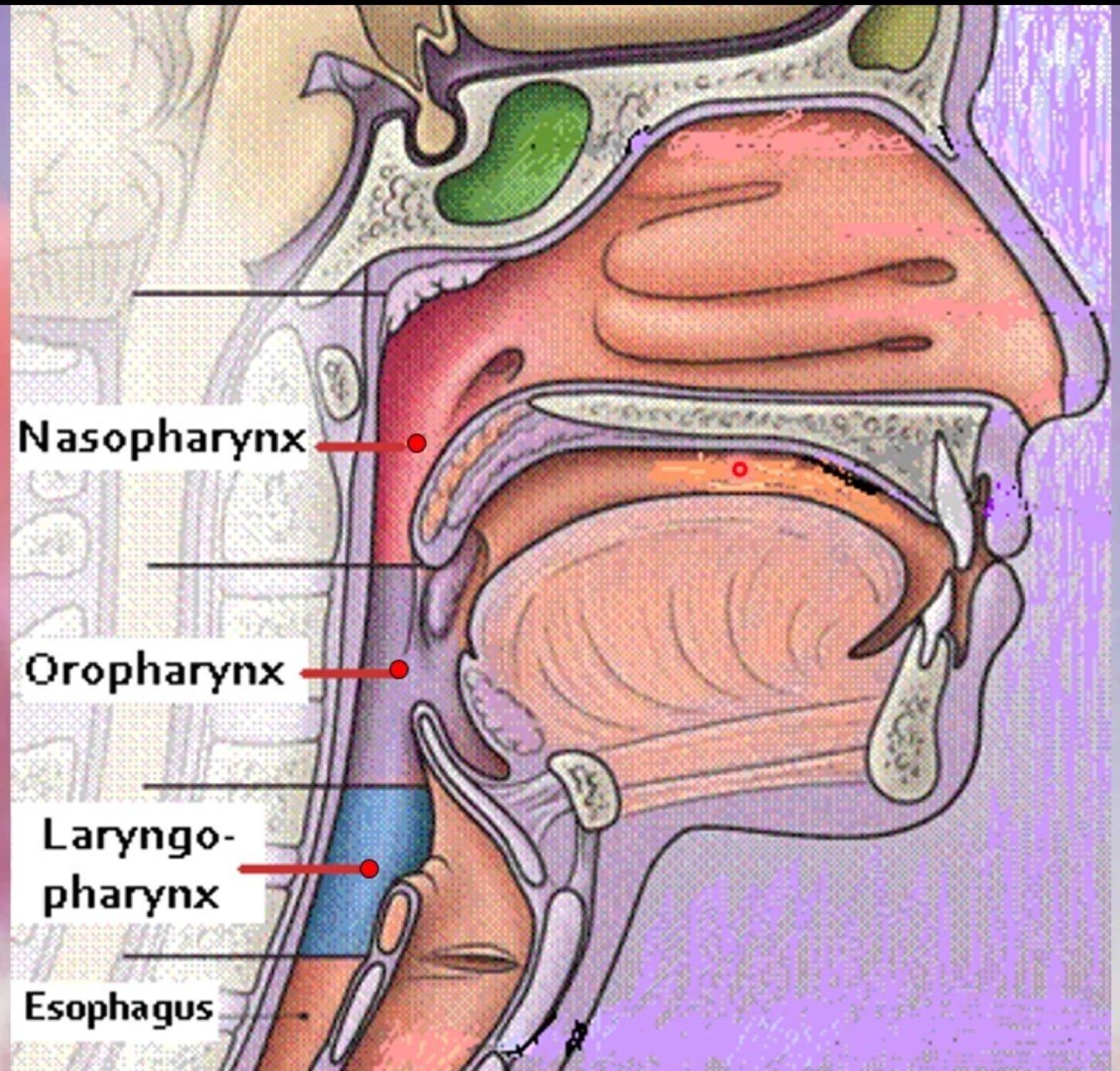
The structures passing upwards deep to the lower border of the inferior constrictor are:

- The recurrent laryngeal nerve.
- The inferior laryngeal vessels.



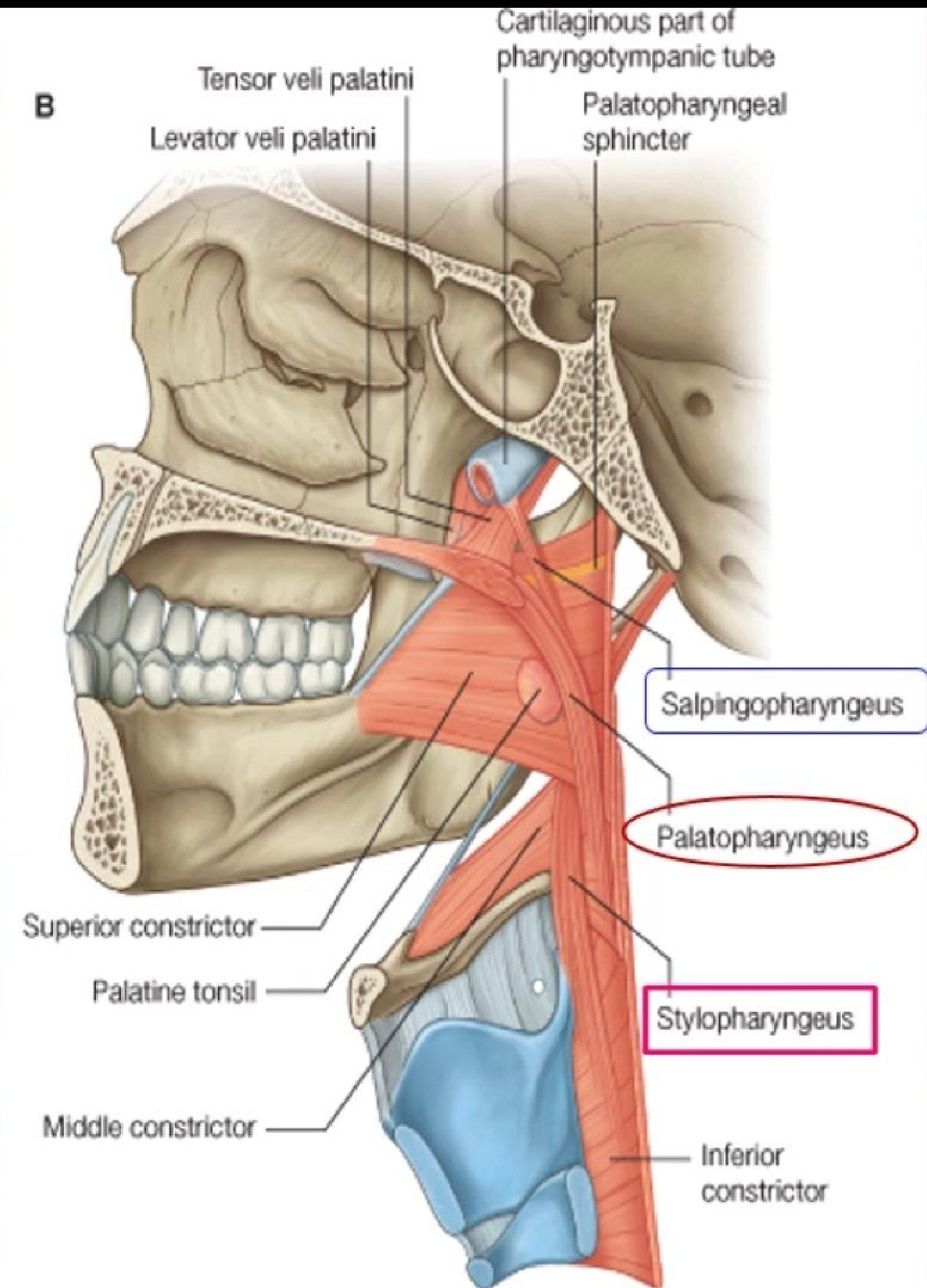
- Pharynx is divided into three parts:

- **Nasopharynx.**
- **Oropharynx.**
- **Laryngopharynx.**



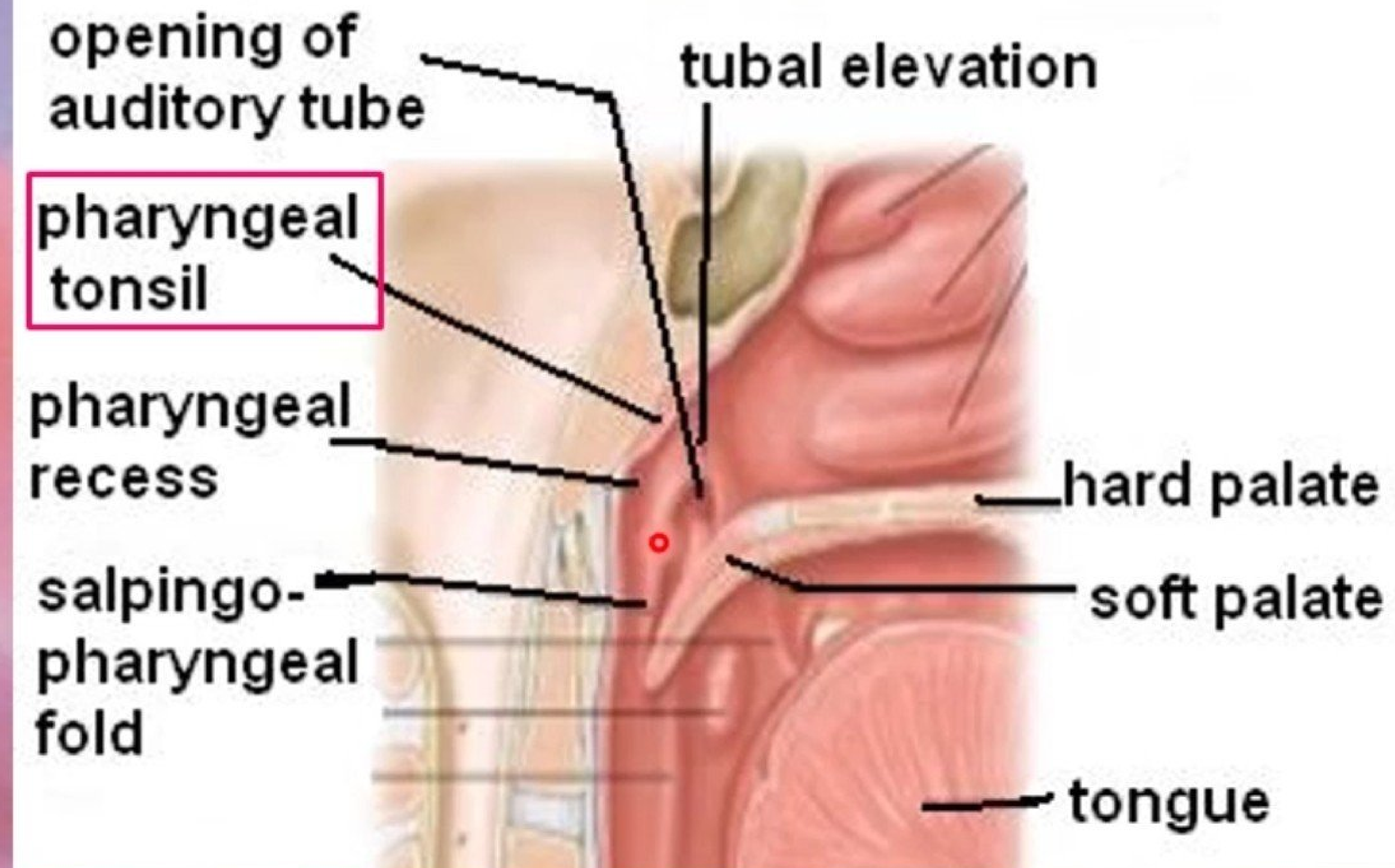
Longitudinal Muscles

- Three in number:
 - **Stylopharyngeus**
 - **Salpingopharyngeus**
 - **Palatopharyngeus**
- Function:
 - Elevate the larynx & pharynx during swallowing.



Nasopharynx

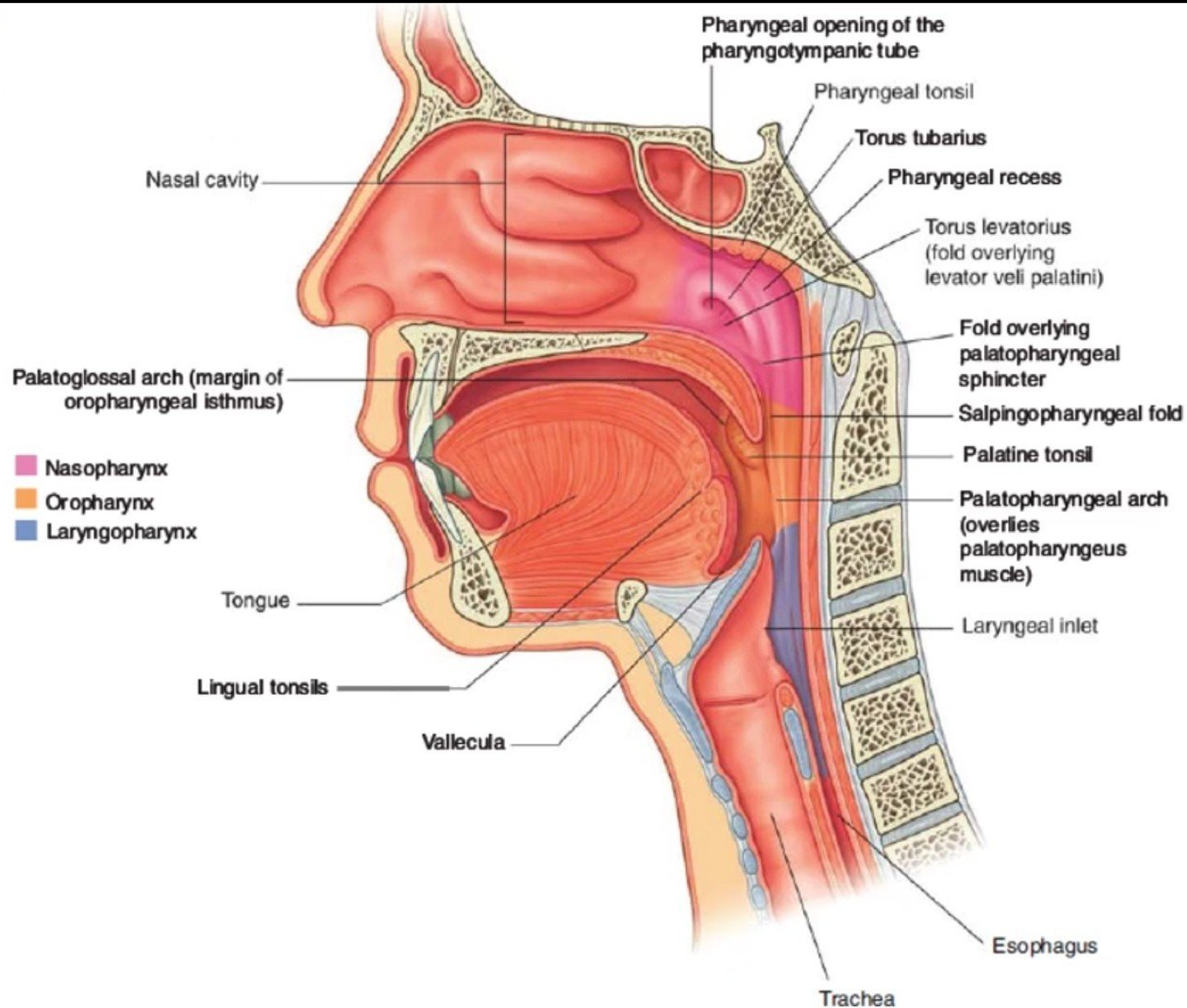
- Extends from the base of skull to the soft palate.
- communicates with the nasal cavity through **posterior nasal apertures**.
- **Pharyngeal tonsils** (Adenoides) present in the submucosa covering the roof.
- Lateral wall shows:
 - **Opening of auditory tube.**
 - **Tubal elevation** (produced by posterior margin of the auditory tube).
 - **Tubal tonsil.**
 - **Pharyngeal recess.**
 - **Salpingopharyngeal fold** (raised by salpingo-pharyngeus muscle).



The pharyngeal recess

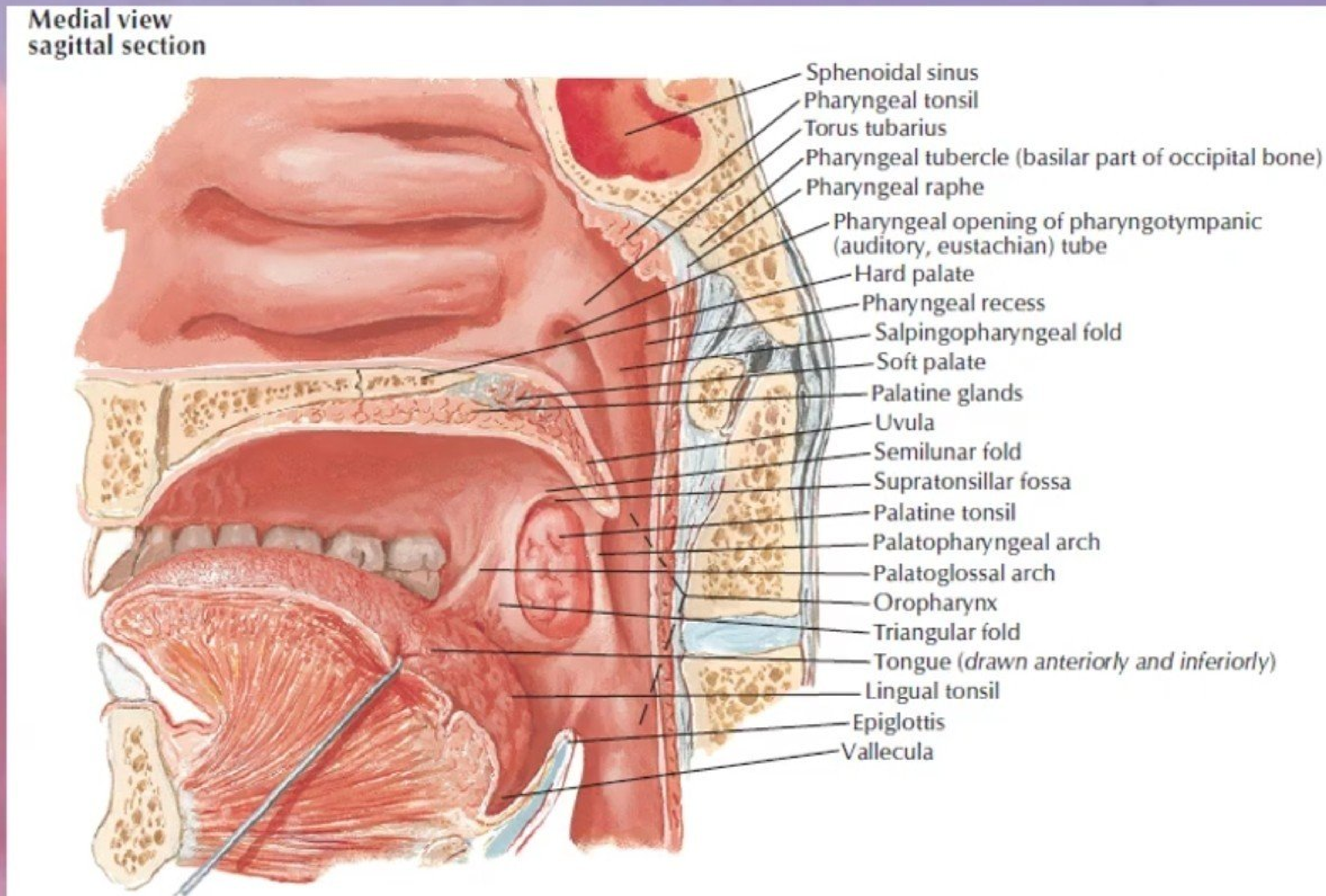
(fossa of Rosenmüller):

- It is a narrow vertical gutter behind the opening of the auditory tube, resulting from the angular attachment of the pharyngobasilar fascia to the base of the skull in front of the carotid canal.
- A catheter missing the tubal orifice and introduced into the recess may perforate the fascia and enter the internal carotid artery, which here lies against the wall of the pharynx.



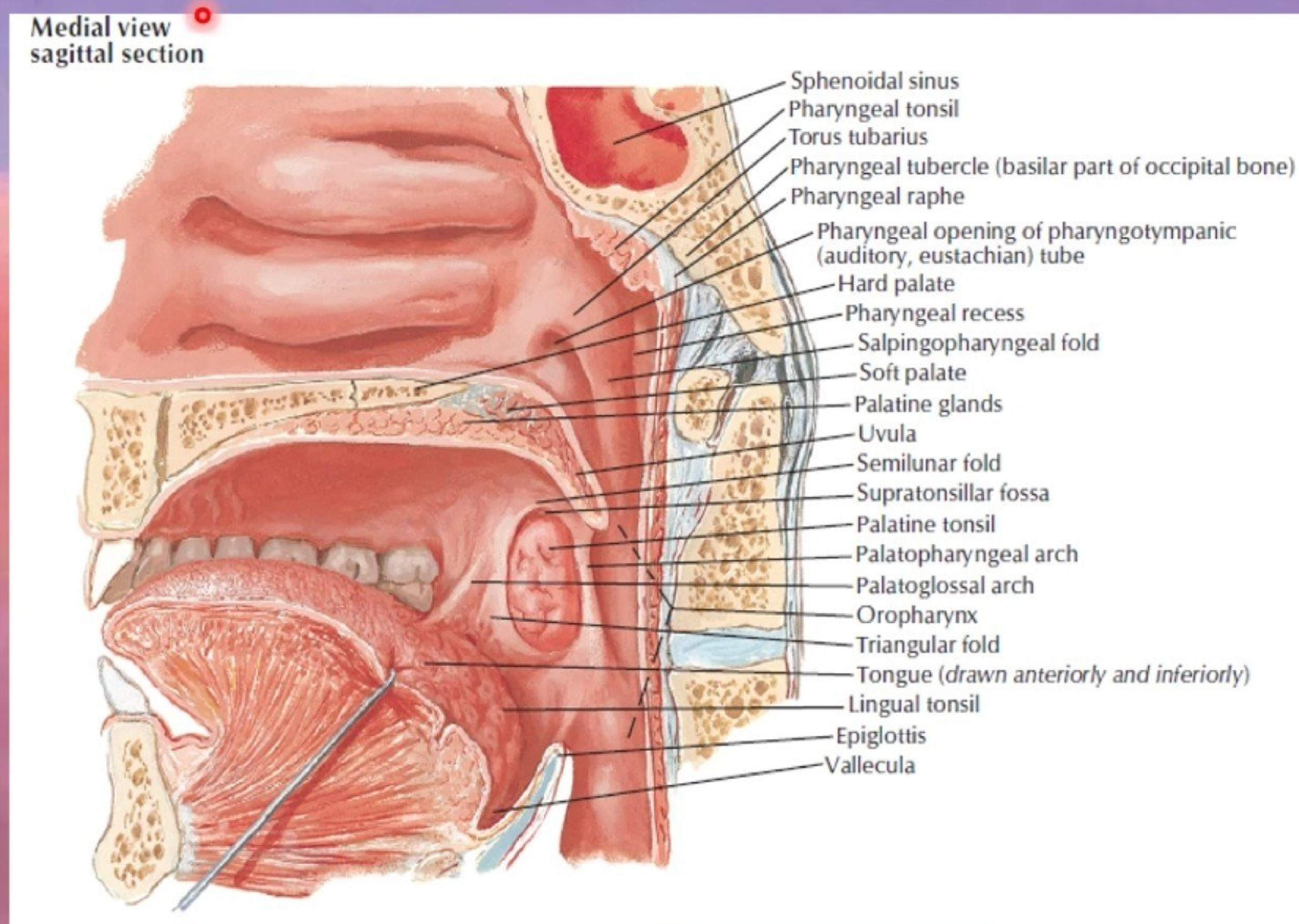
Oropharynx

- Lies behind the mouth, communicates with the oral cavity through the **oropharyngeal isthmus**
- Extends from **soft palate** to **upper border of epiglottis**.
- **Lateral wall shows:**
 - **Palatopharyngeal fold.**
 - **Palatoglossal fold**
 - **Palatine tonsil** located between them in a depression called the 'tonsillar fossa'.

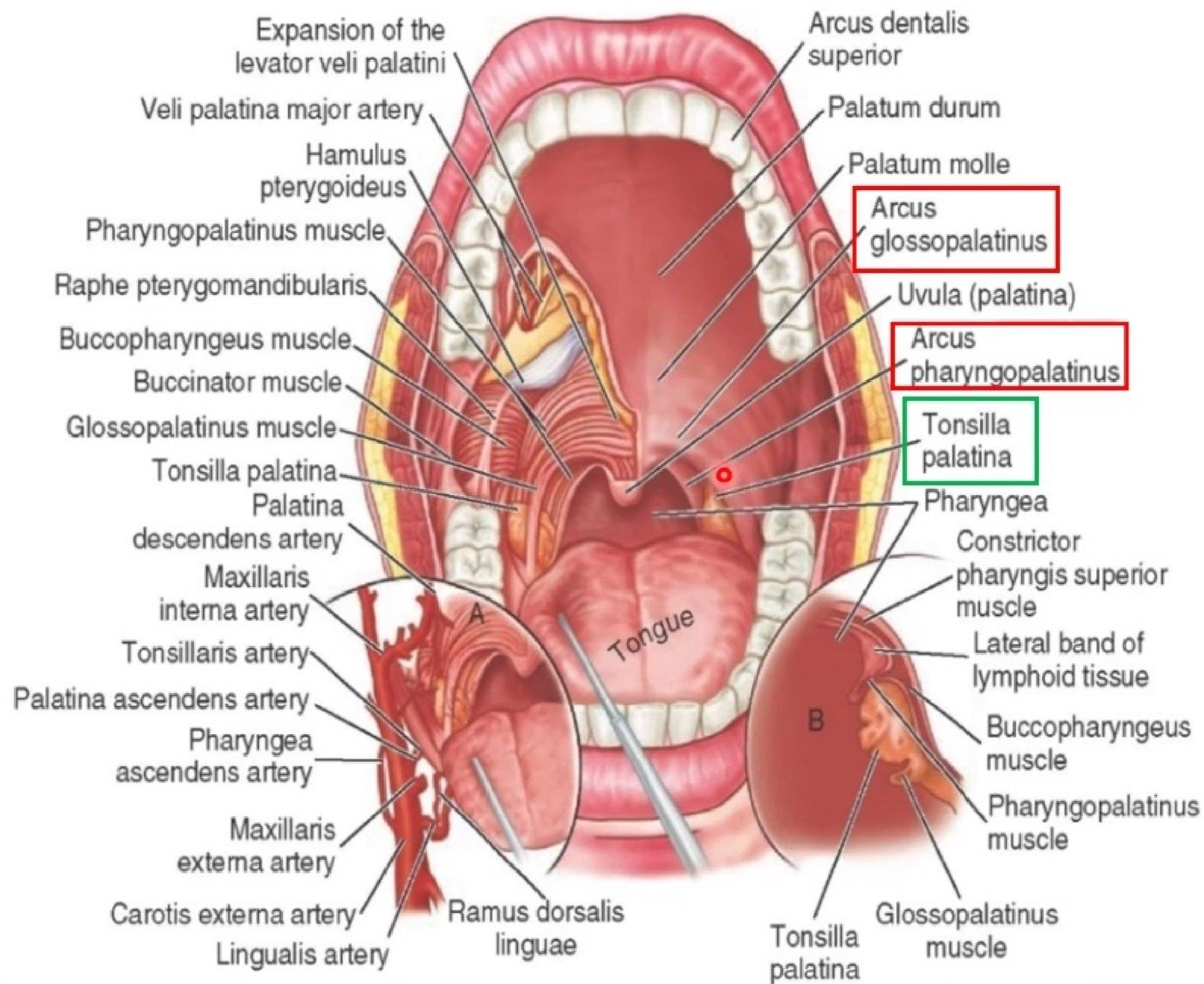


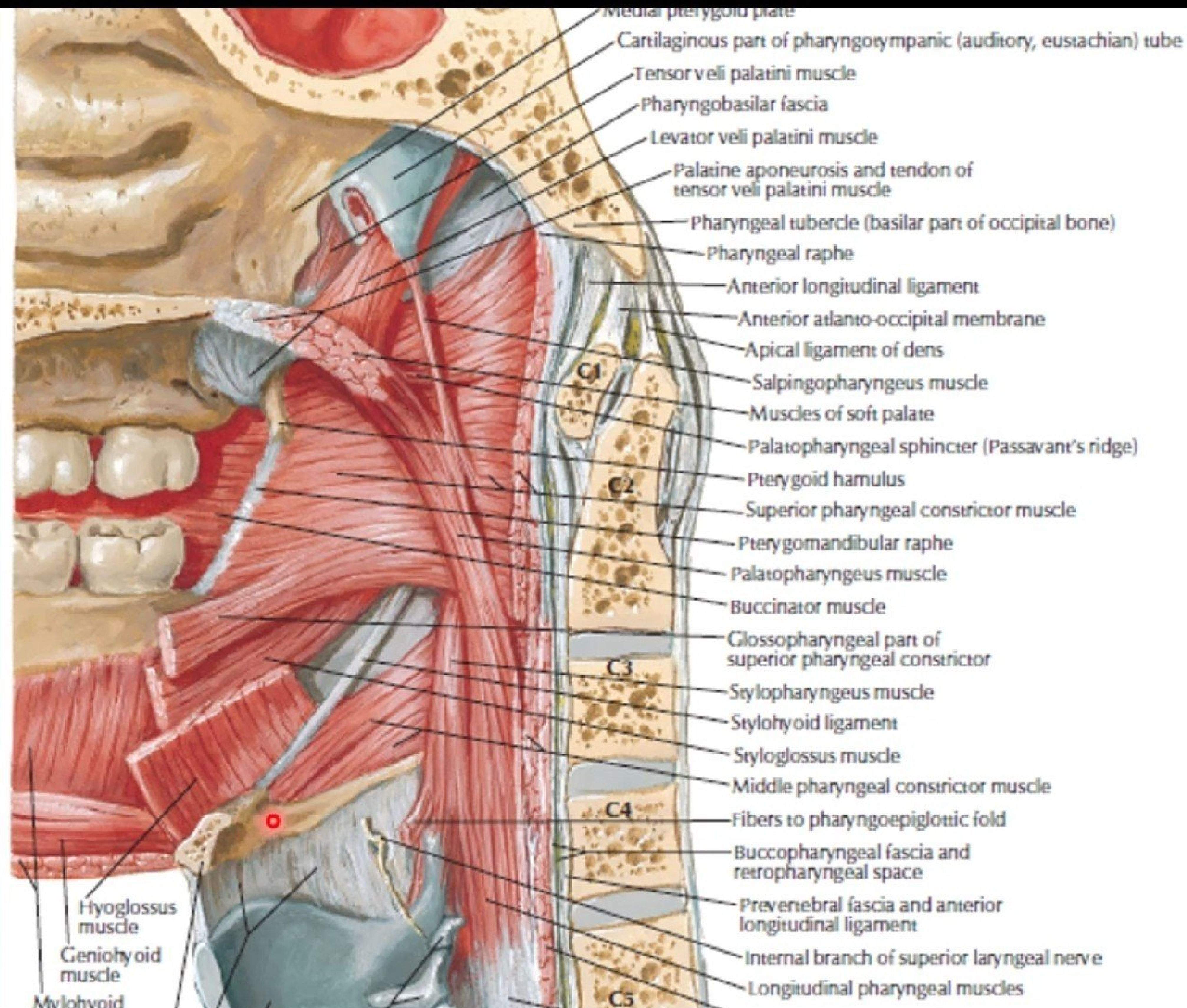
The palatine tonsil (tonsils)

- It is a large collection of lymphoid tissue which projects into the oropharynx.
- It is found in the *tonsillar fossa* between the palatopharyngeal fold behind and the palatoglossal fold in front.
- The glossopharyngeal nerve crosses the lower part of the floor of the *tonsillar*.
- The superior constrictor separates this surface from the facial artery and two of its branches, the ascending palatine and tonsillar.



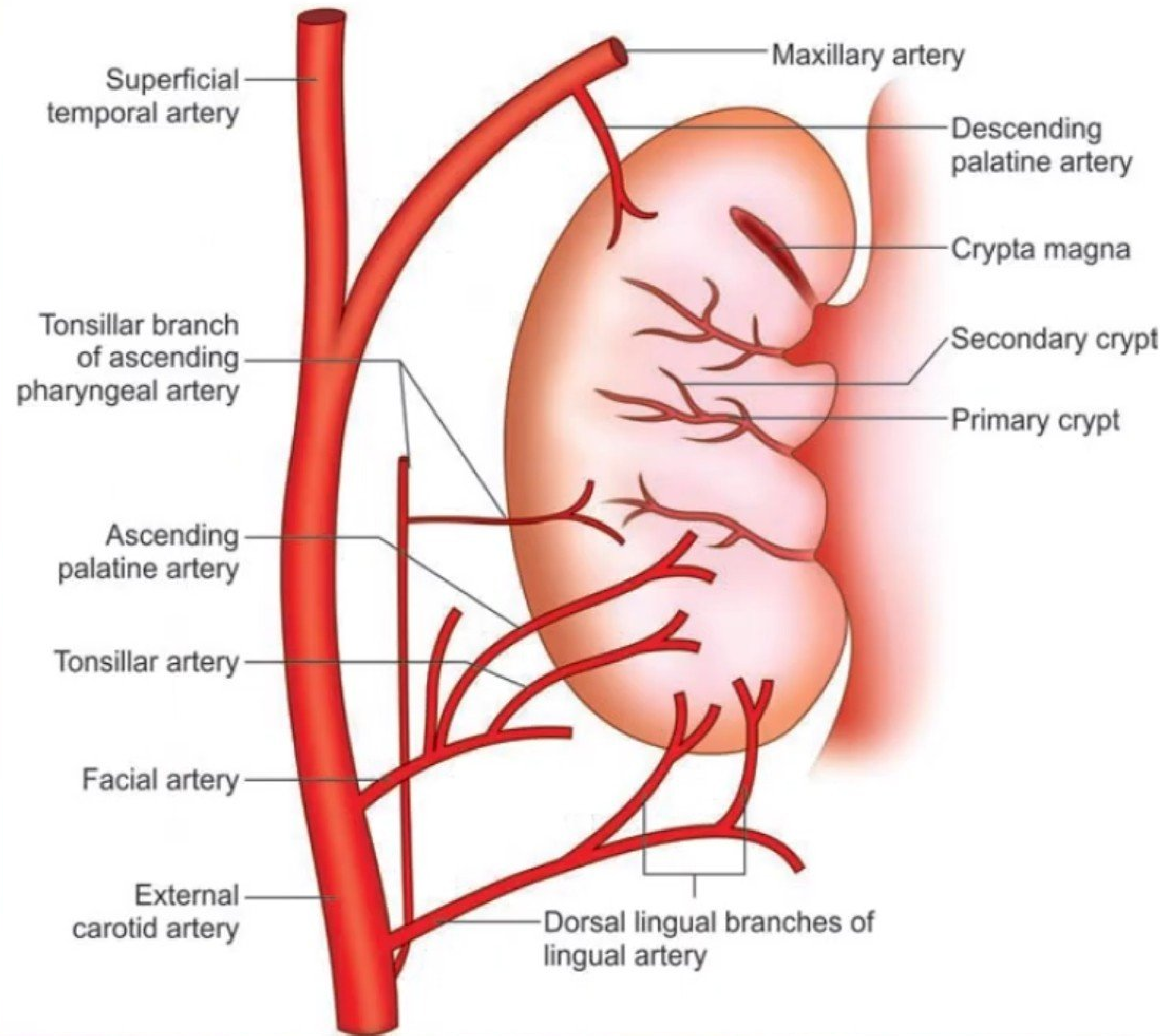
- The internal carotid artery is about 2.5 cm posterolateral to the tonsil.

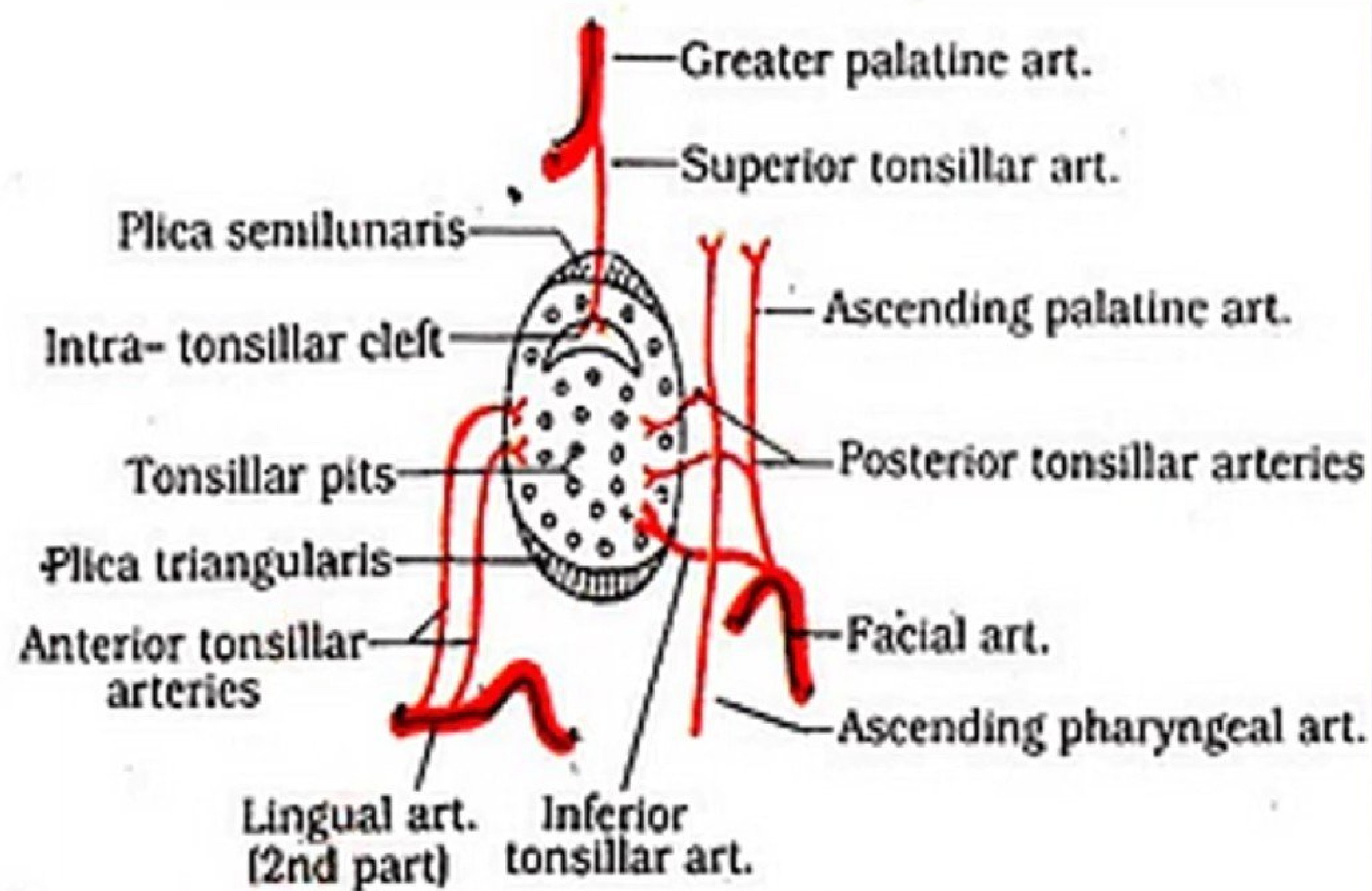
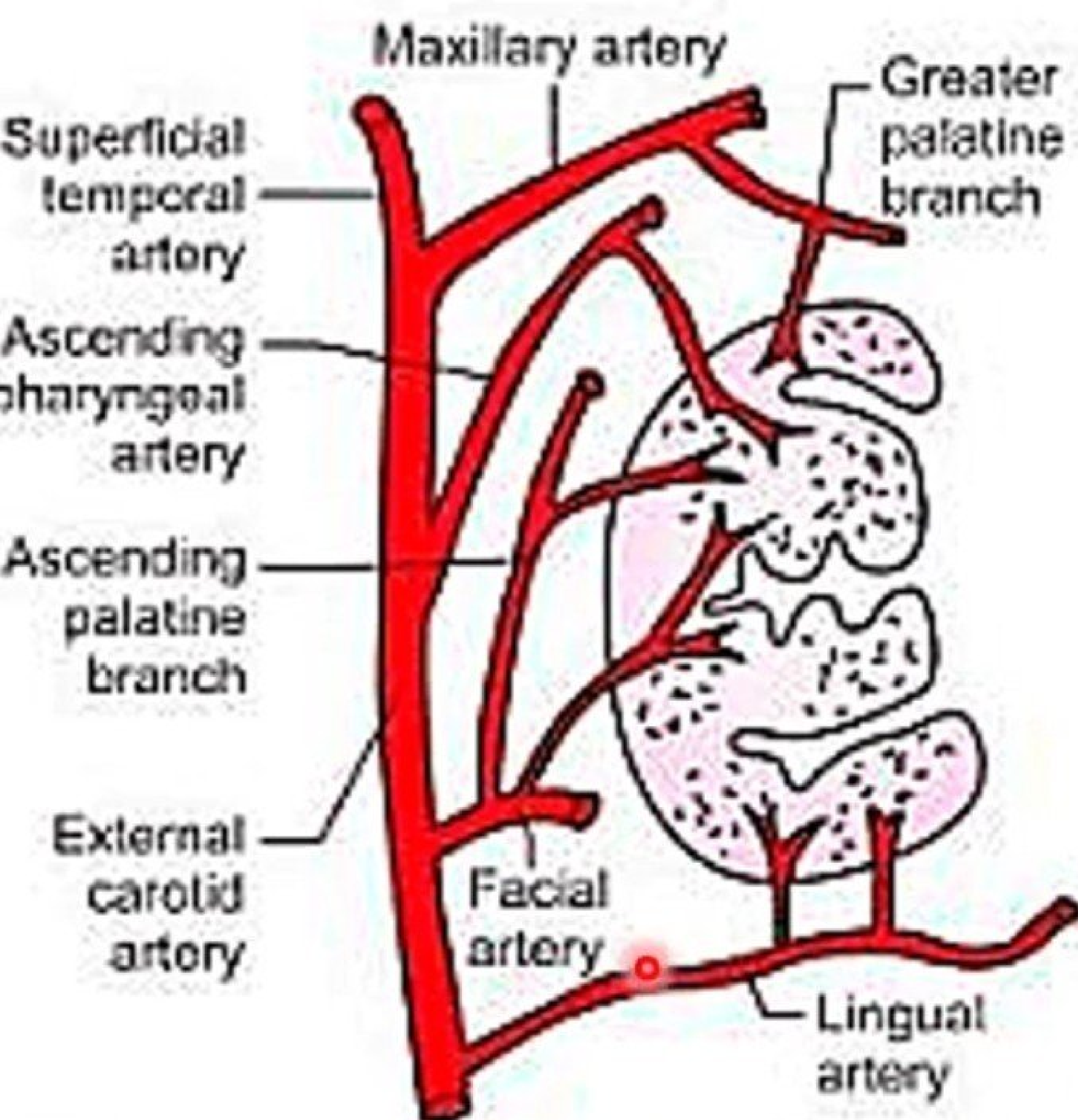




Blood supply:

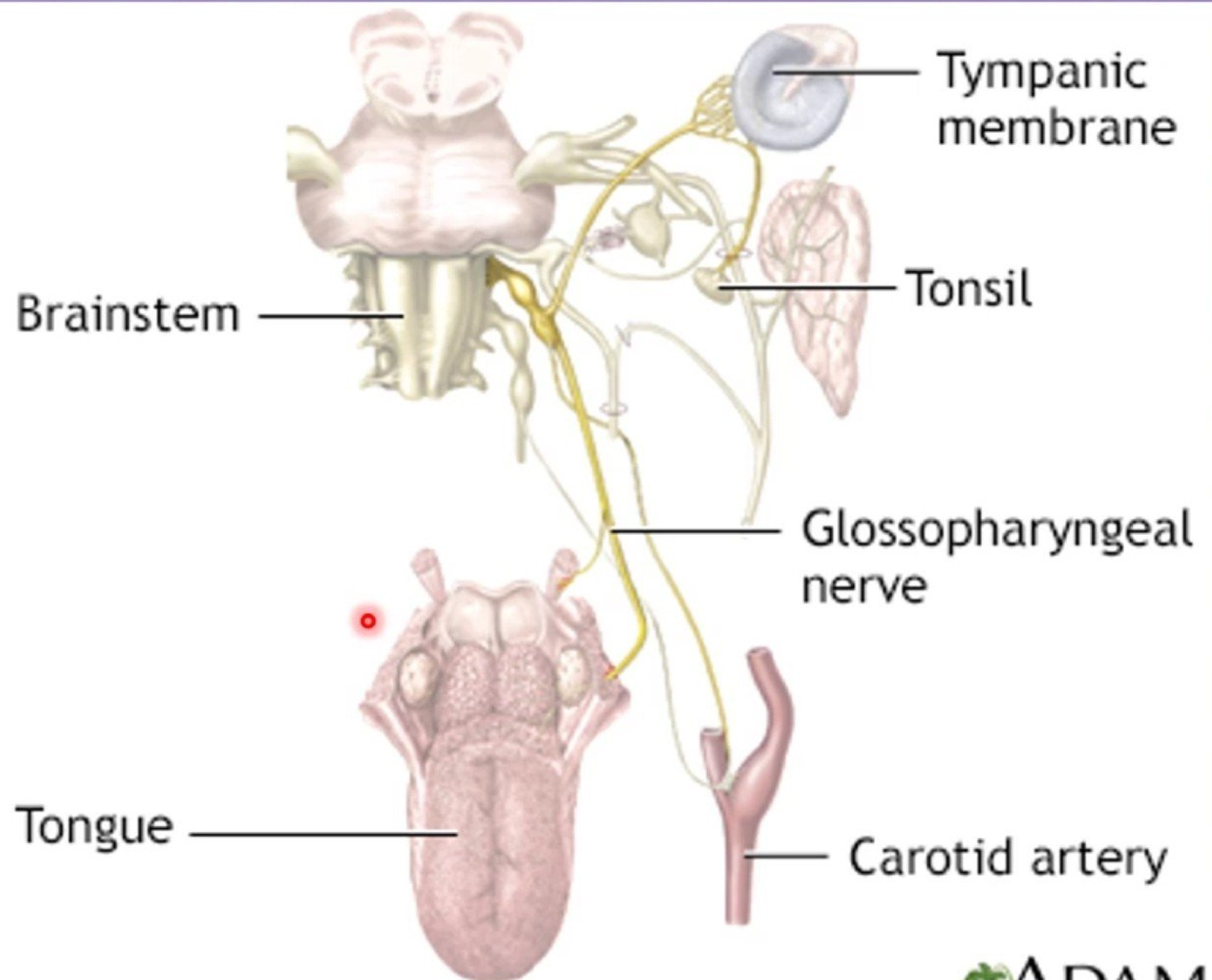
- The tonsillar branch of **the facial artery** forms the main arterial supply.
- It enters the tonsil by piercing the superior constrictor.
- There are smaller contributions from the lingual, ascending pharyngeal and ascending and greater palatine vessels.



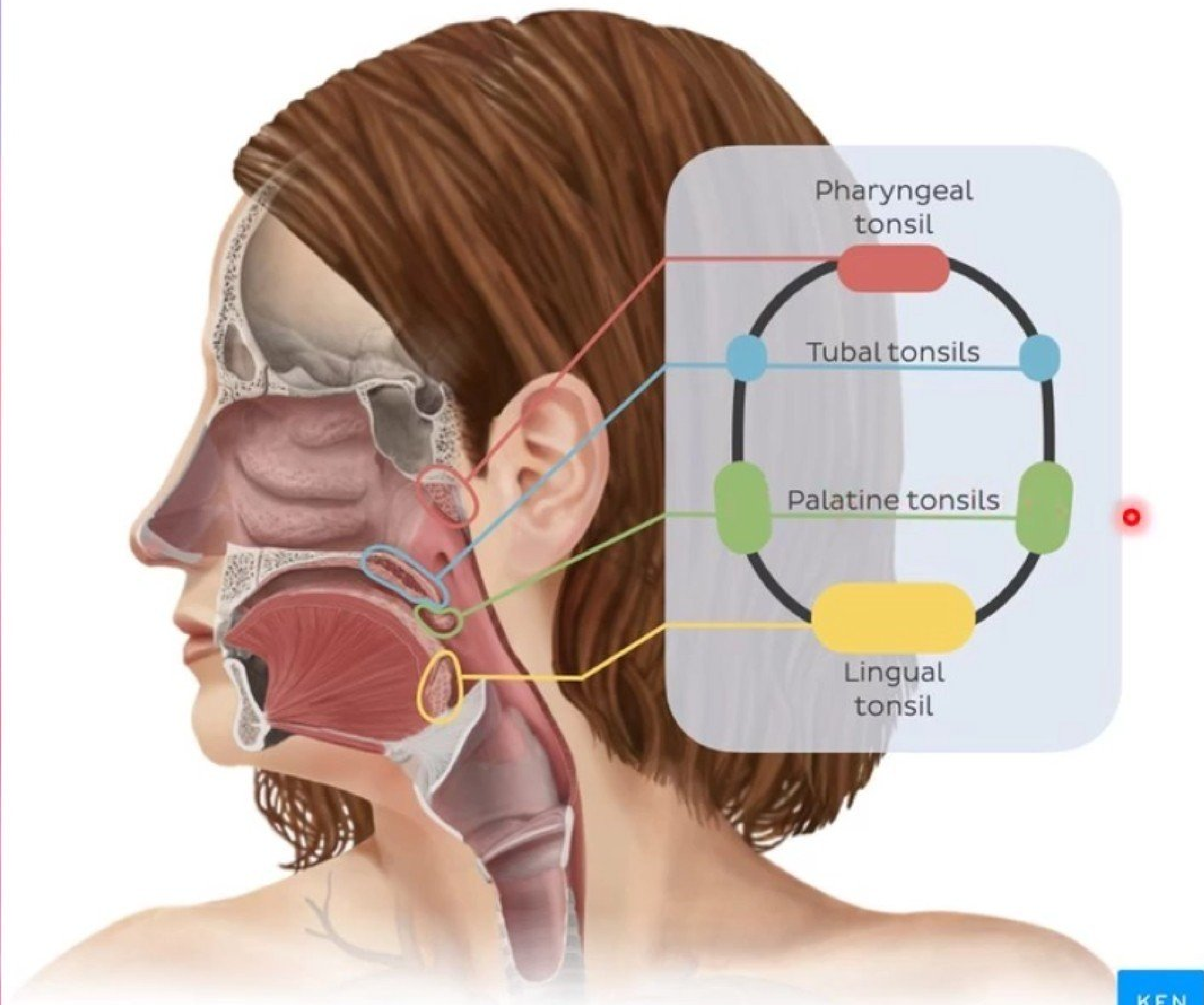


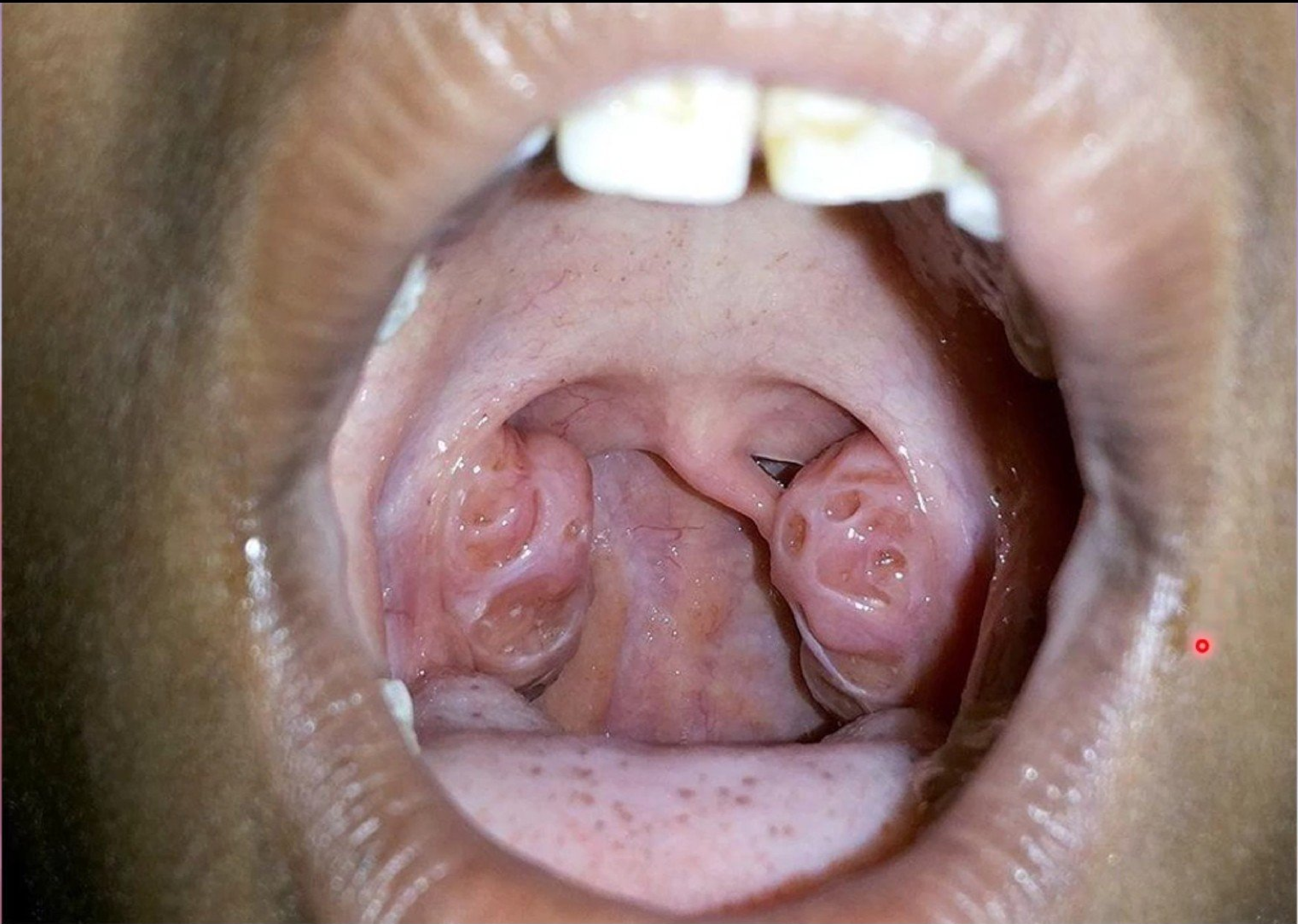
Nerve supply.

- The mucous membrane overlying the tonsil is supplied mainly by the **tonsillar branch of the glossopharyngeal nerve**, and to a small extent by the lesser palatine nerves.
- The **glossopharyngeal nerve** also supplies the middle ear, through its tympanic branch, and tonsillitis may cause referred pain in the ear.



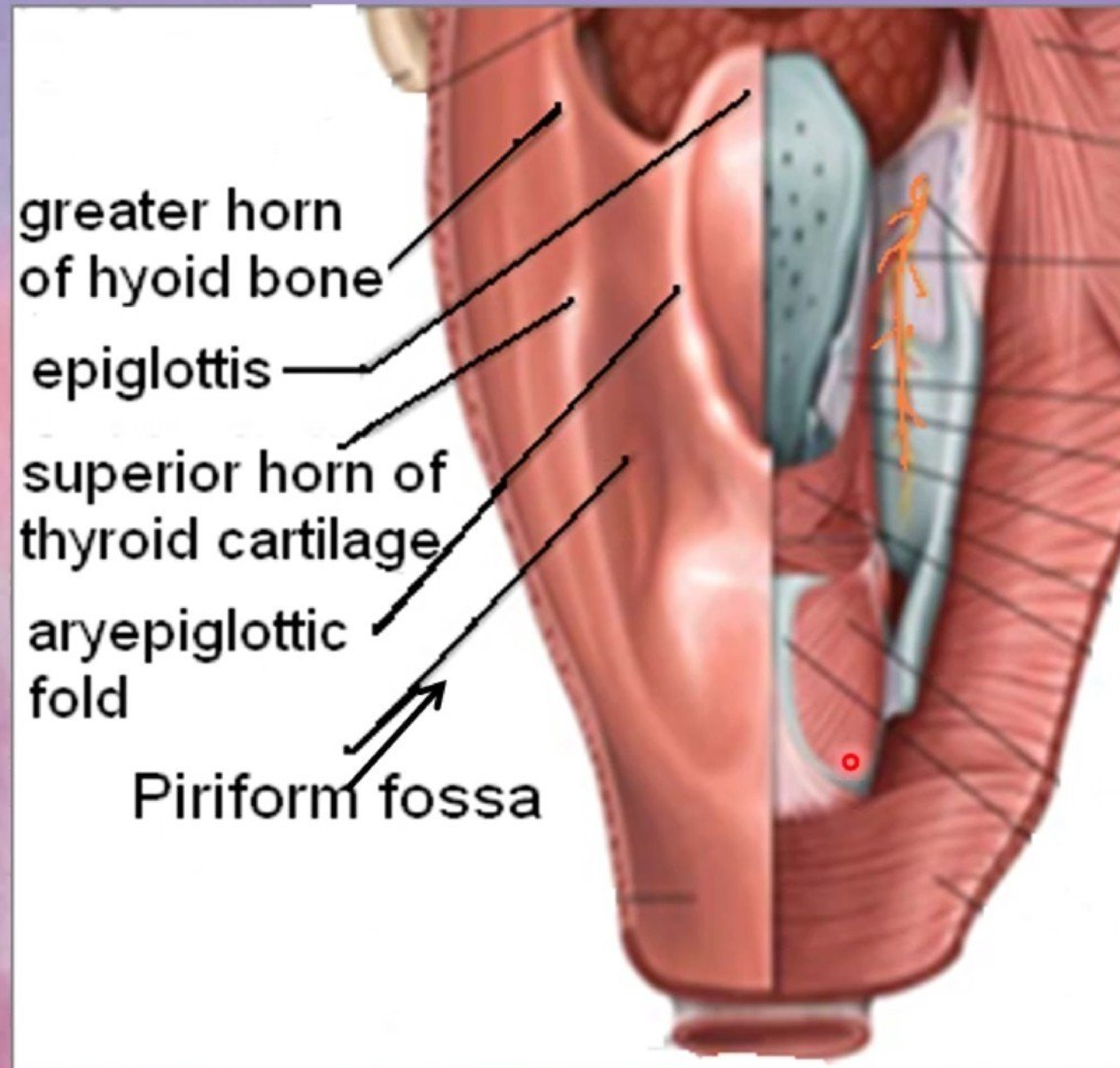
The **palatine, lingual, pharyngeal** and **tubal tonsils** collectively form an interrupted circle of **lymphoid tissue** (*Waldeyer's ring*) at the upper end of the respiratory and alimentary tracts.





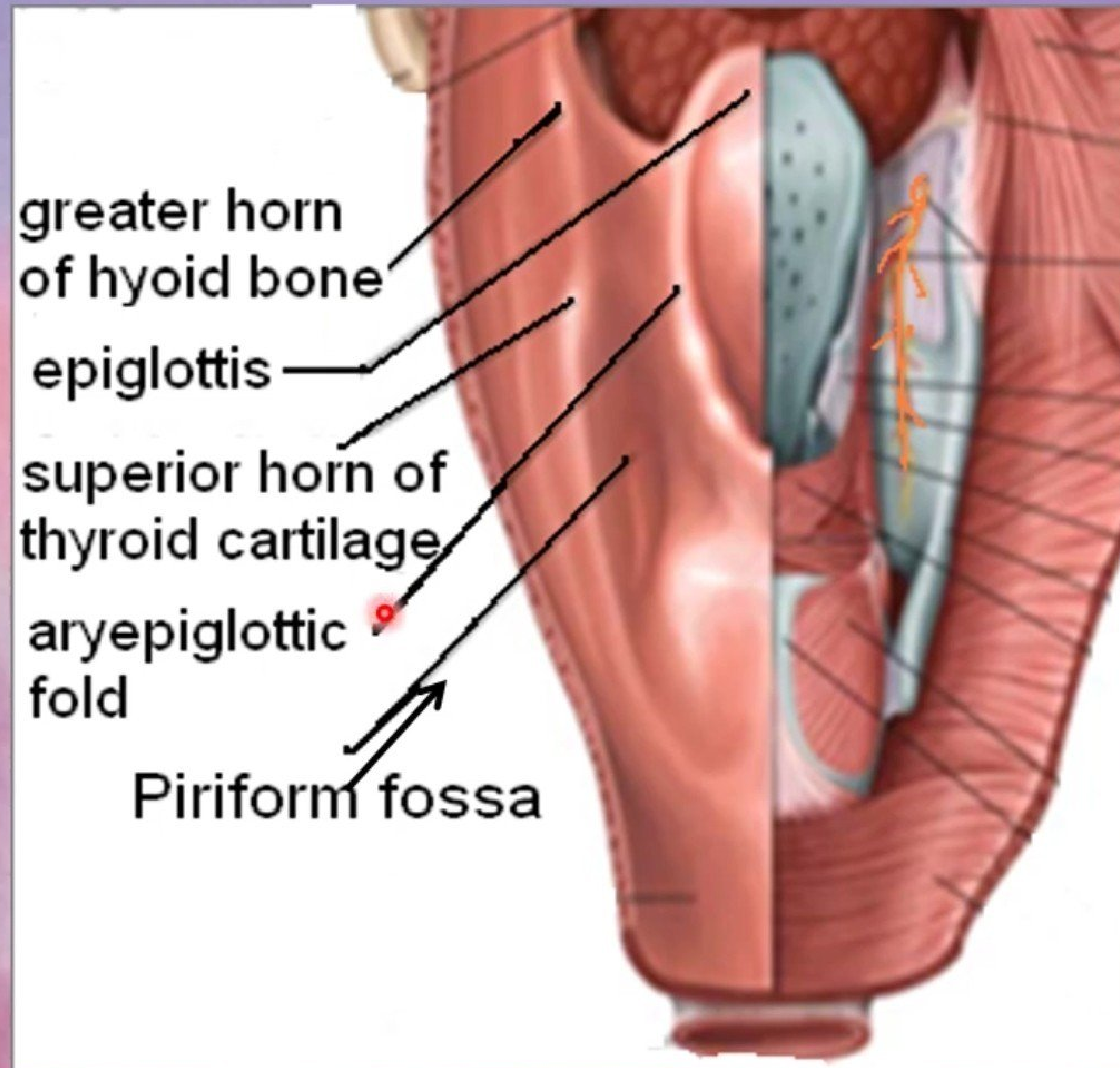
Laryngopharynx

- Lies behind the laryngeal inlet & the posterior surface of larynx.
- communicates with the larynx through the laryngeal inlet
- Extends from upper border of epiglottis to lower border of cricoid cartilage.
- A small depression situated on either side of the laryngeal inlet is called '**Piriform Fossa**'.
- It is a common site for the lodging of foreign bodies.
- Branches of internal laryngeal & recurrent laryngeal nerves lie deep to the mucous membrane of the fossa and are vulnerable to injury during removal of a foreign body.

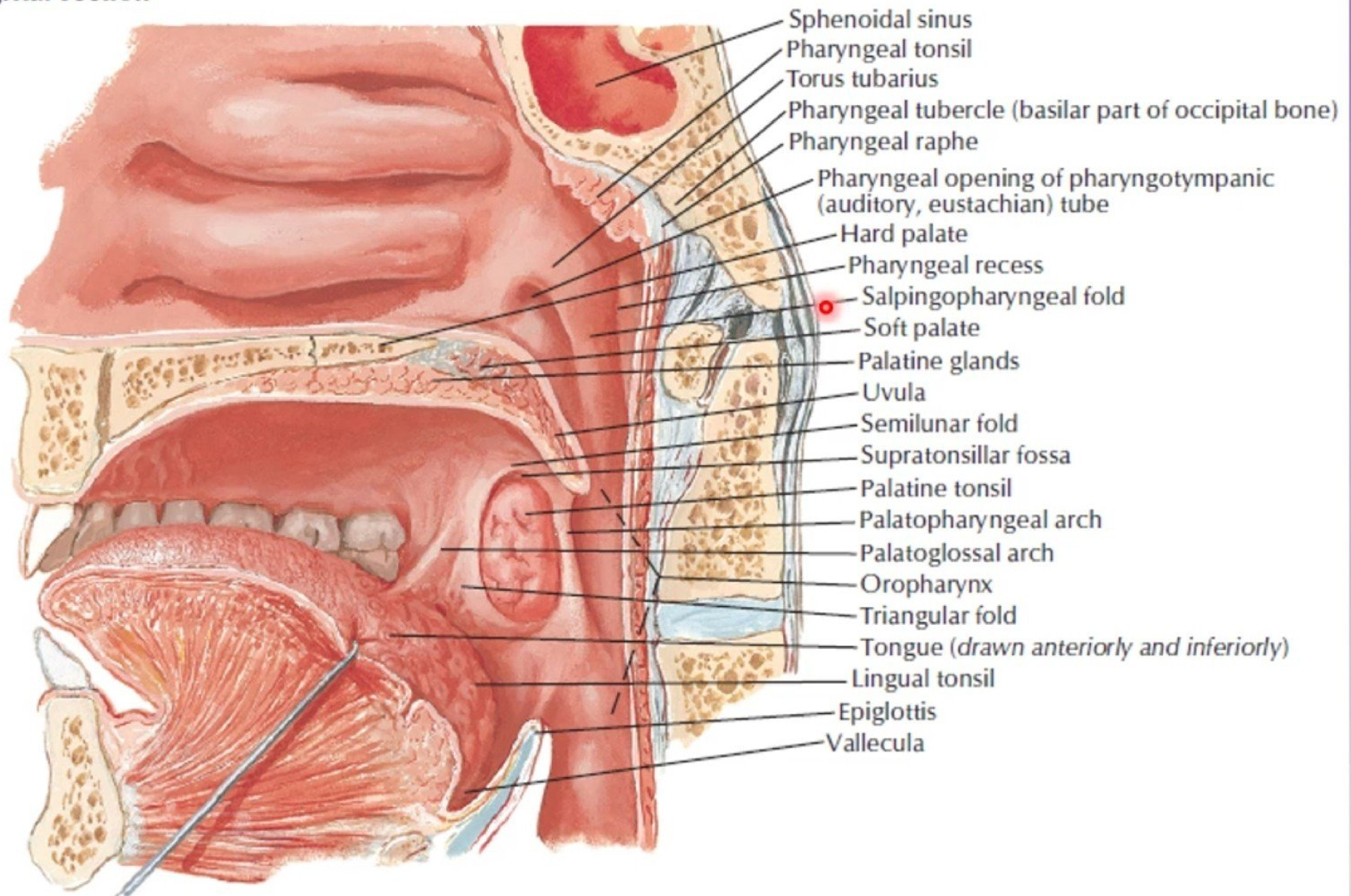


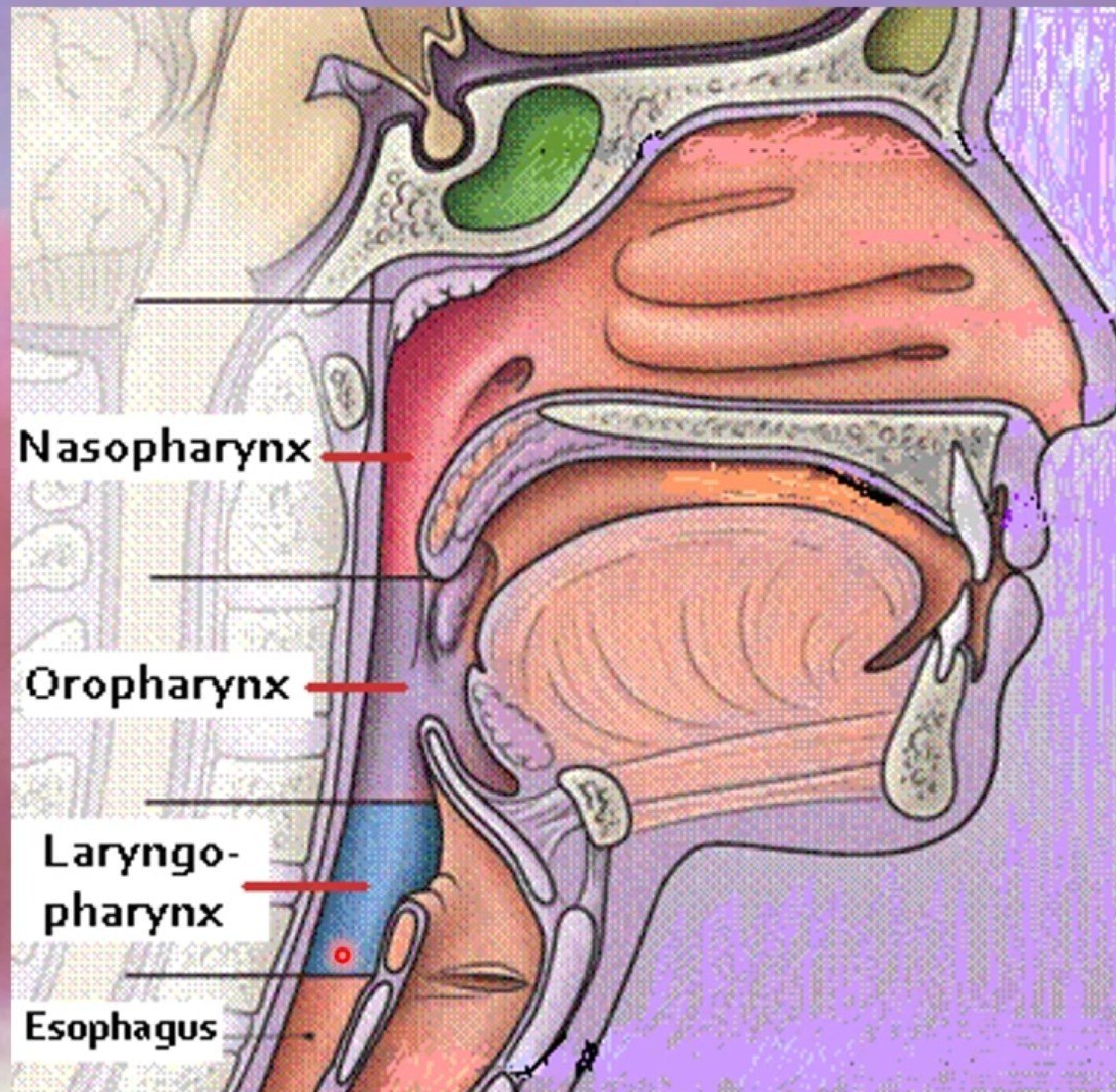
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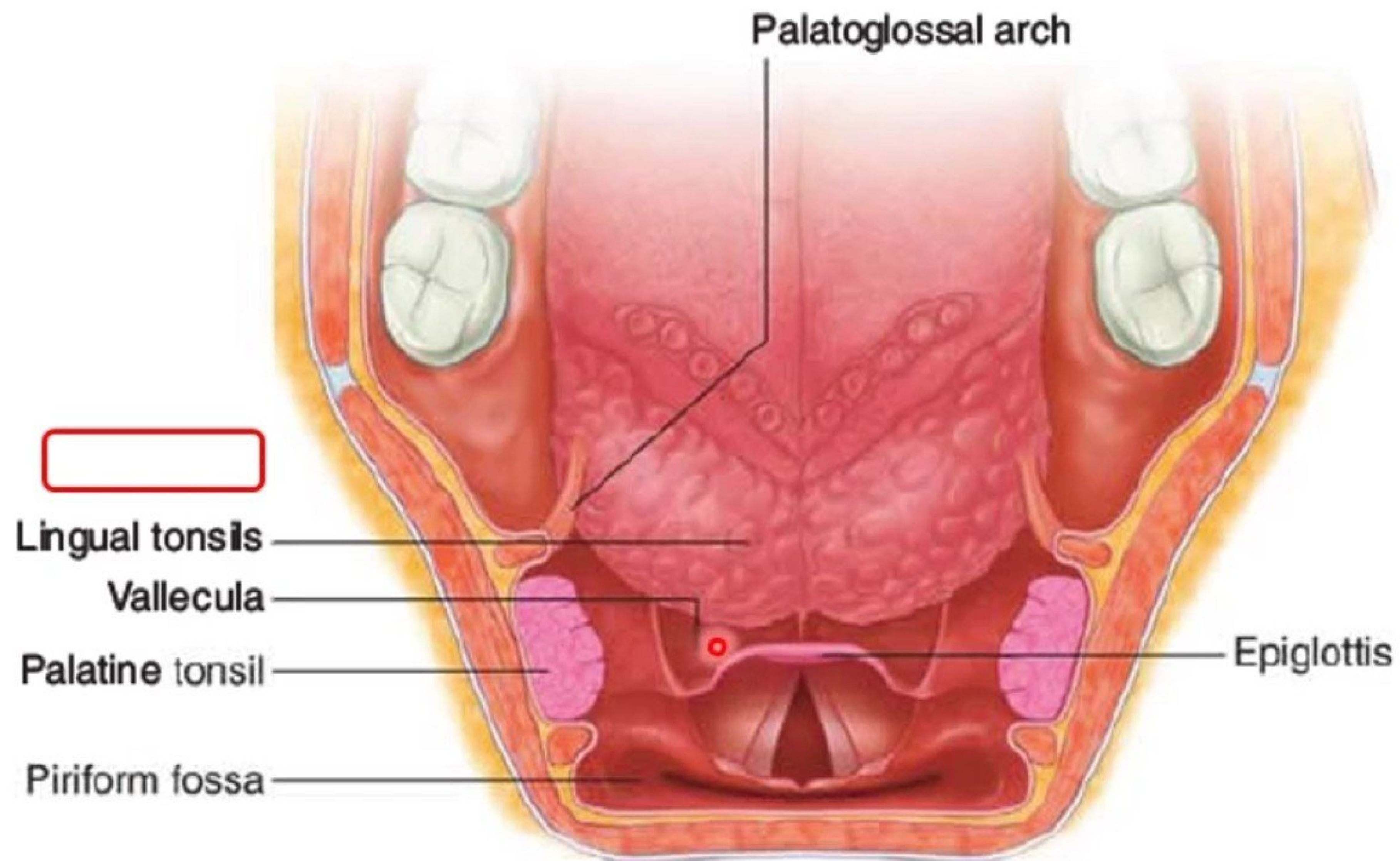
Medial view
sagittal section





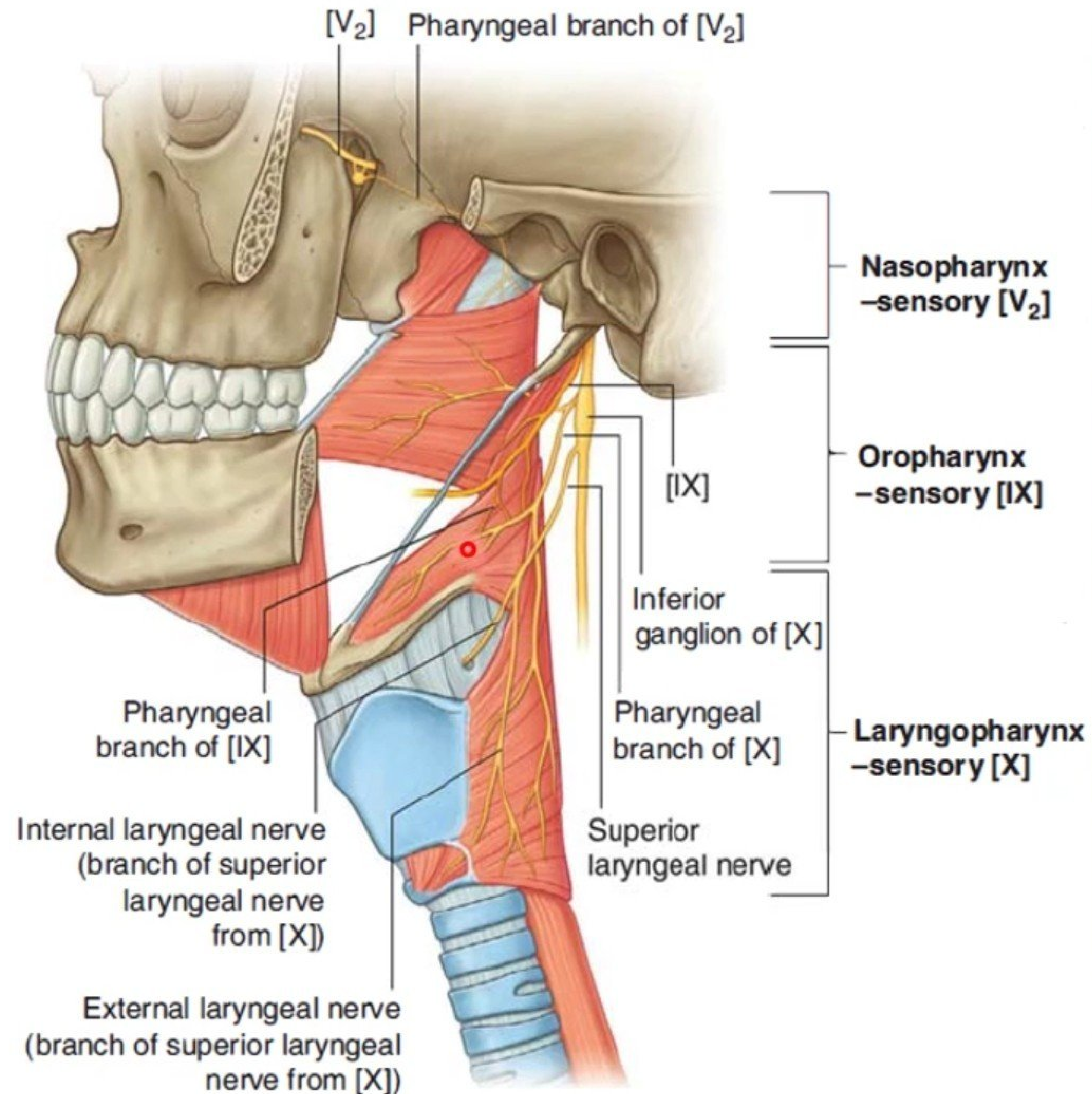
The valleculae

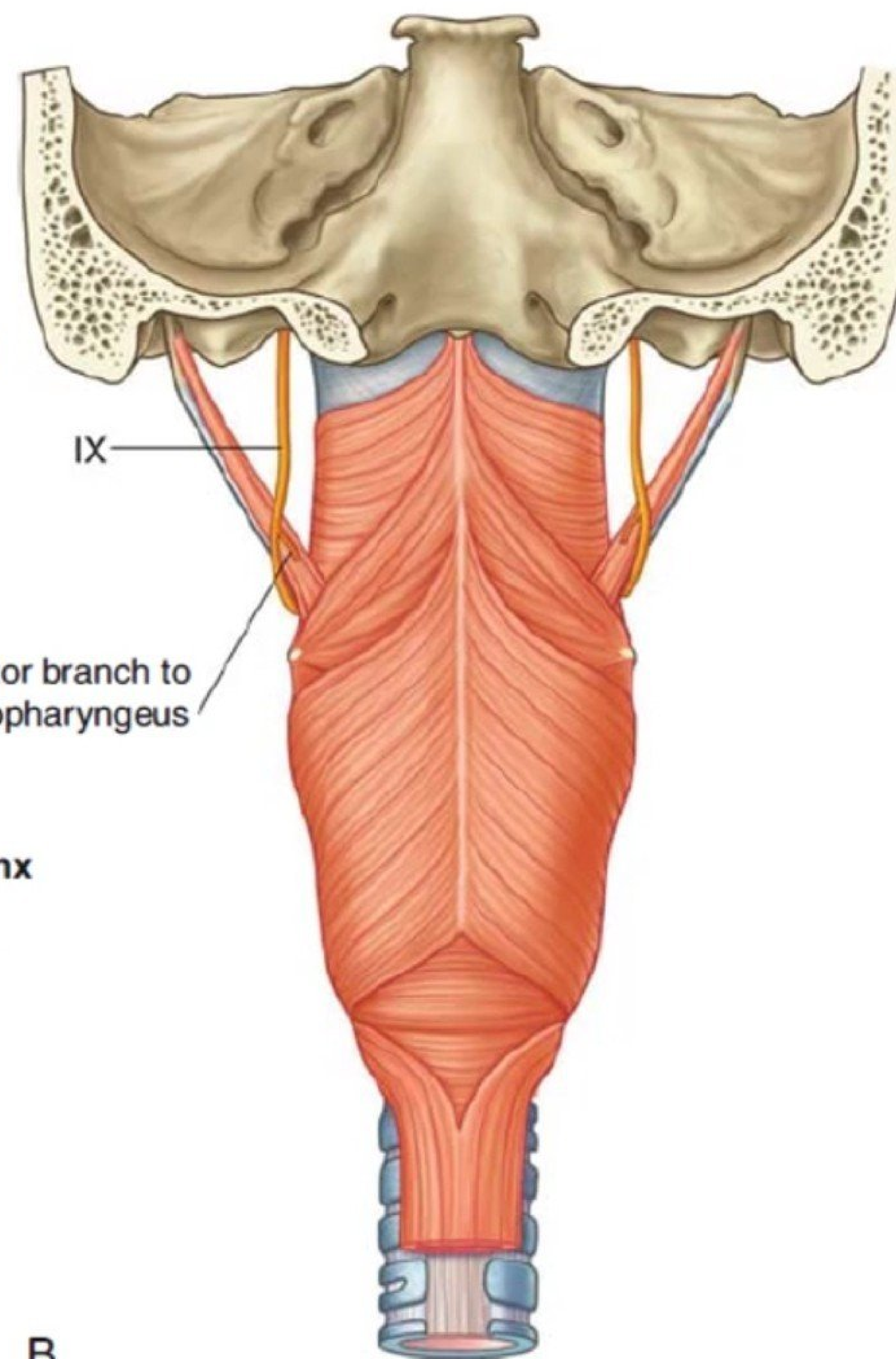
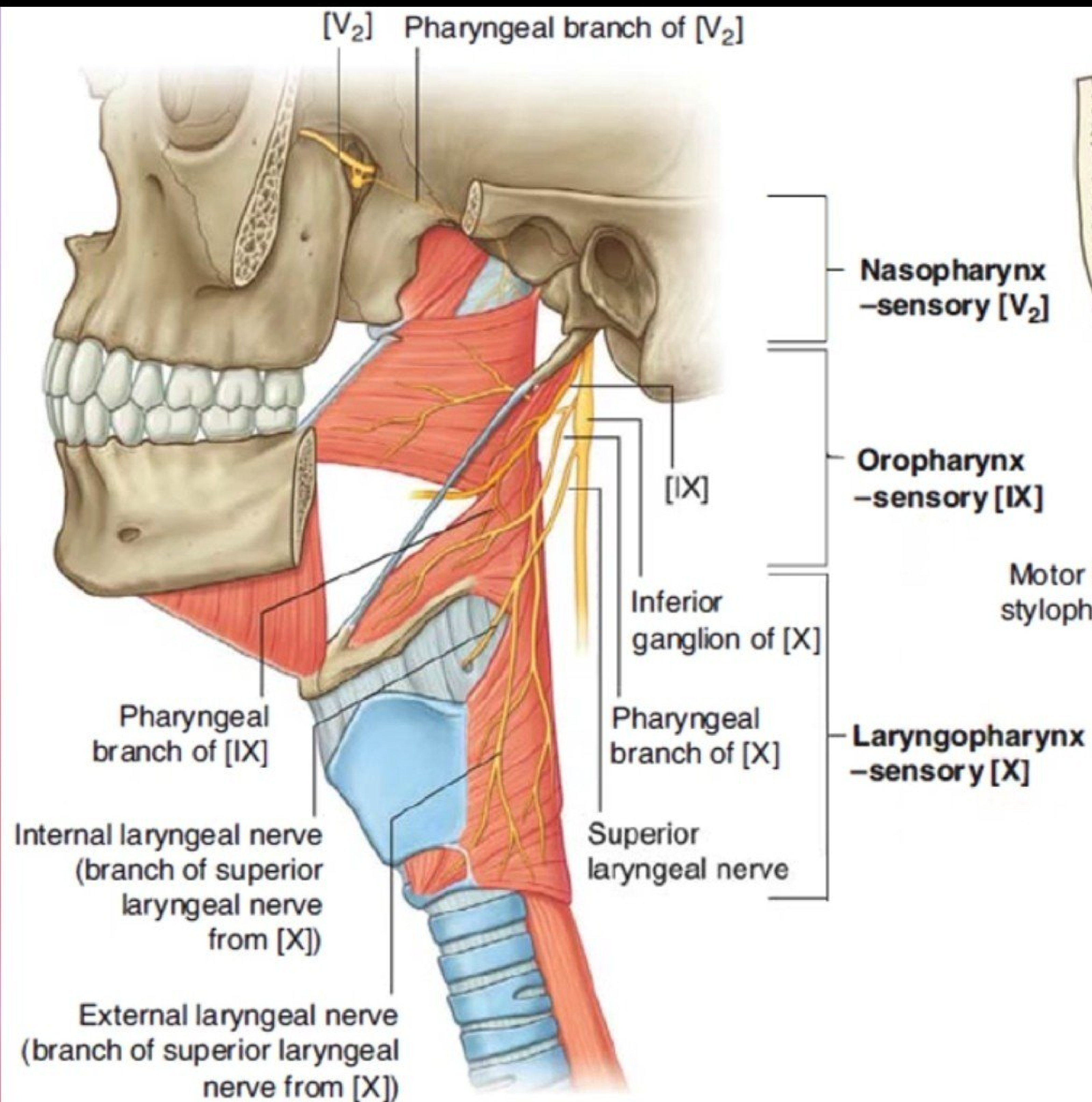
- They lie between the epiglottis and the posterior surface of the tongue.
- They are shallow fossae separated by the median glossoepiglottic fold and limited inferolaterally by the lateral glossoepiglottic folds.
- The nerve supply of the mucosa by the internal laryngeal nerve.
- If any thing that lodges in the vallecula that stimulate a coughing reflex to dislodge it.



Nerve Supply

- **Sensory: (mucosa)**
- **Nasopharynx: Maxillary nerve**
(supplied by the pharyngeal branch of the maxillary nerve through the pterygopalatine ganglion)
- **Oropharynx: Glossopharyngeal nerve**
(but the internal laryngeal nerve supplies the valleculae and the lesser palatine nerves (maxillary) contribute to the supply of the tonsillar mucosa)
- **Laryngopharynx: Vagus nerve**
(The internal laryngeal nerve)
- **Motor :**
 - All the muscles of pharynx are supplied by the pharyngeal plexus.
 - Except ; the **Stylopharyngeus** is supplied by the glossopharyngeal nerve.
- **Cricopharyngeus** is also supplied by the **recurrent** and **external laryngeal nerves**.



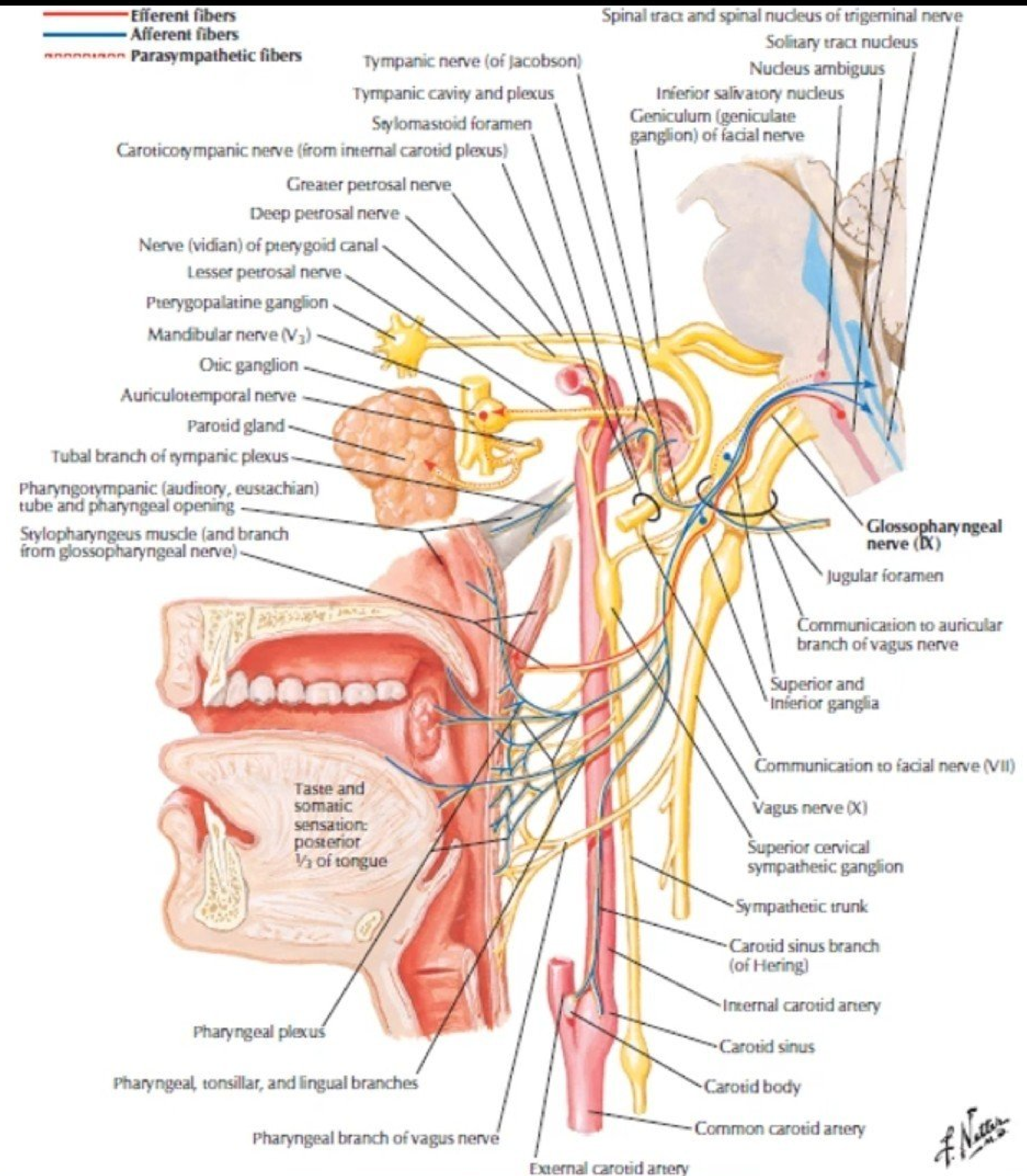


A

B

The pharyngeal plexus

- It lies on the posterolateral wall of the pharynx, mainly over the middle constrictor.
- It is formed by the union of:
 - The pharyngeal branches from the vagus.
 - The pharyngeal branches from glossopharyngeal nerves and
 - The external laryngeal nerve branches from the superior laryngeal branch of the vagus nerve.
 - the cervical sympathetic nerves.
- The glossopharyngeal component is afferent.
- the pharyngeal fibres of the vagus carry motor fibres (derived from the cranial part of the accessory nerve).
- The sympathetic fibres are vasoconstrictor.
- **Note:**
 - The pharyngeal branches from the vagus is the major motor nerve supply of the pharynx.
 - All muscles of the pharynx are innervated by vagus nerve (through the pharyngeal plexus) except ; the Stylopharyngeus is supplied by the glossopharyngeal nerve.



Arterial supply:

from branches of the following arteries:

1) Facial artery supply pharynx by:

- Ascending pharyngeal artery.
- Ascending palatine artery.
- Tonsillar artery.

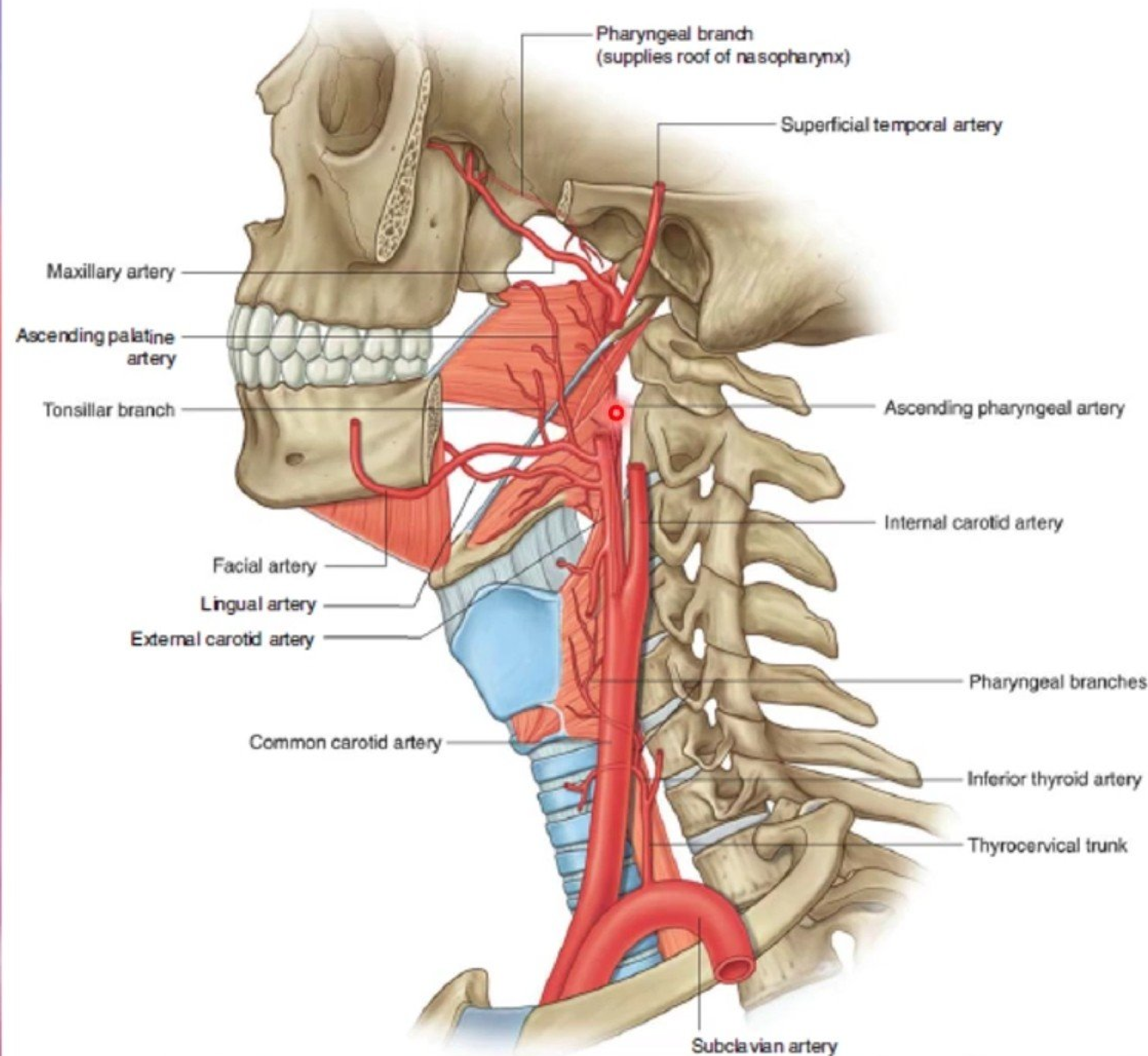
2) Maxillary artery supply pharynx by:

- greater palatine artery.
- Pharyngeal artery.

3) lingual artery.

4) Superior and inferior laryngeal arteries.

5) Pharyngeal artery (from inferior thyroid artery).

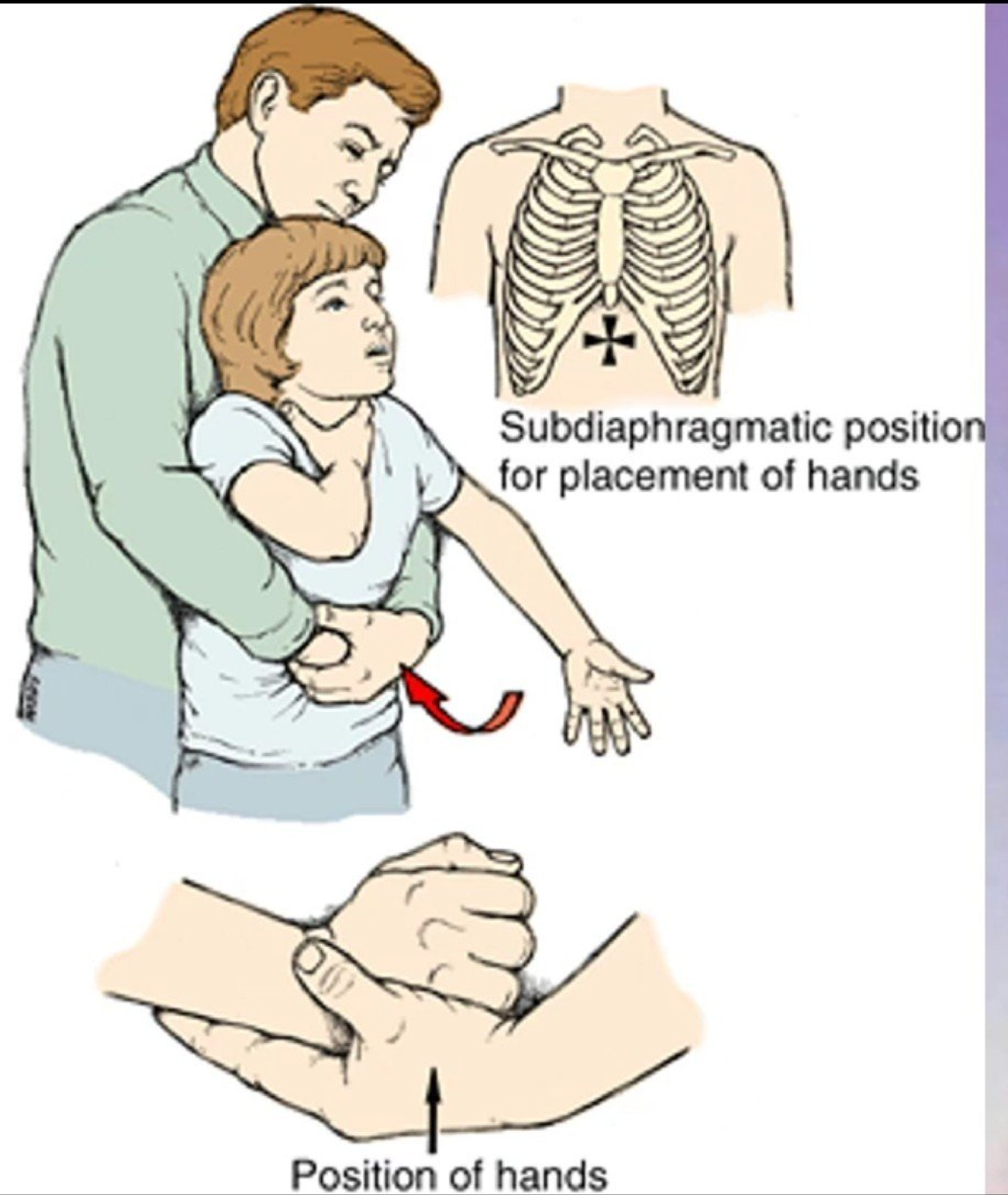


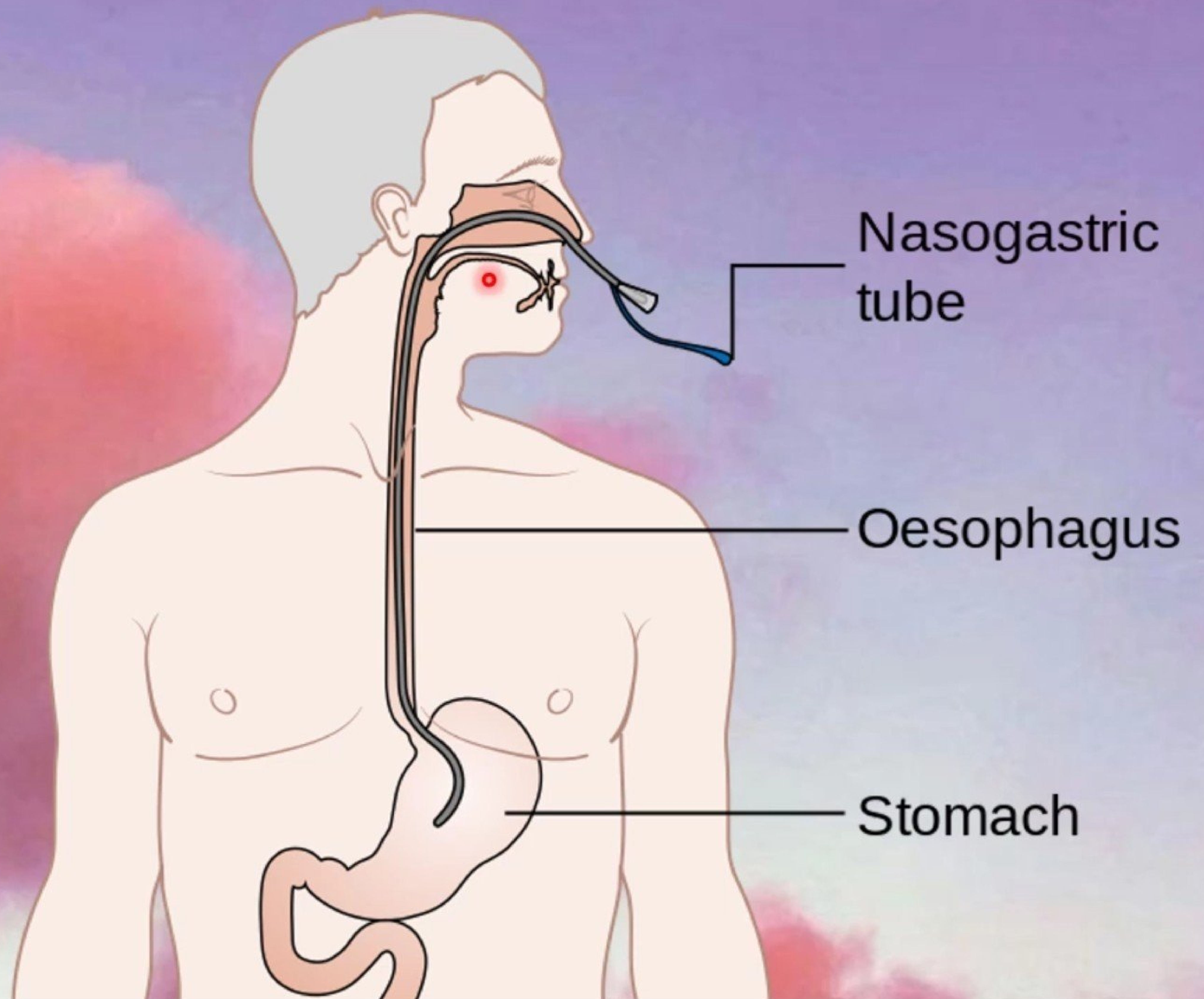
Foreign Bodies in the Laryngopharynx

Sinus Tract from
the Piriform
Recess



Heimlich maneuver





*Thank You
& Good Luck*